

The rise of large genomes and biological databases has opened entirely new opportunities for computational genomics research. However, the core task of this field, classifying the pathogenicity of genes, is held back by the heterogeneity of biological data and complex interactions between genes. Furthermore, experimental validation pathways are time-consuming and costly, making this work extremely difficult to advance.

Conventional genetic variation classification schemes generally rely on single-source databases or a single statistical method. They cannot capture the complex multi-layered associations among genetic variation, gene function, and interaction networks, and they also overlook the molecular interaction dimension. These limitations ultimately lead to reduced prediction accuracy and insufficient biological interpretability.

To address the unsolved problems put forward in the foregoing text, this study proposes a novel bioinformatics computing solution: at the underlying layer, it integrates data from four categories of biological databases including the genetic variation database, builds a preprocessing pipeline to complete two core transformations, adopts a hybrid framework combining machine learning and graph neural networks as its core, and conducts analysis using two types of information simultaneously.

The gene pathogenicity prediction framework independently developed for this study delivers excellent prediction efficiency for both pathogenic and non-pathogenic genes. The core driver of its performance improvement is the introduction of graph learning, which optimizes prediction accuracy at the data interaction dimension.

Preamble of the chapter

The continuous iterative advancement of high-throughput sequencing technology, paired with the gradual popularization of large-scale biological datasets, has completely transformed the development landscape of the computational genomics field. Starting from the basic biological functions of genes, the accurate assessment of gene pathogenicity is the core challenge of current cross-disciplinary research; traditional biological analysis methods rely on isolated datasets and statistical tools, and cannot capture the complex associations between genes and proteins. New technologies such as Artificial Intelligence (AI), Machine Learning (ML), and Graph Neural Networks (GNNs) provide support for the efficient analysis of large-scale genomic data, and this chapter will focus on introducing the core content of AI-driven gene-level pathogenicity prediction frameworks.

1.1 Introduction

Genes are the basic functional genetic units that regulate various biological processes in living organisms. Genetic variations such as mutations and sequence changes can disrupt cellular mechanisms and trigger physiological abnormalities. These variations fall into two categories: pathogenic and benign. Identifying pathogenic genes can help advance understanding of disease mechanisms, optimize diagnostic practices, and support biomedical research.

In recent years, genomic databases such as ClinVar, NCBI Gene, and STRING have expanded rapidly, providing researchers with vast amounts of biological data. However, their large scale and heterogeneity have rendered manual analysis unsustainable, and traditional methods are also unable to model the interaction dependencies between genes and proteins.

To address the research challenges facing the field, machine learning and graph neural networks have demonstrated core value in the bioinformatics domain. The former can

learn complex patterns from high-dimensional biological data, while the latter can model gene associations relying on graph structures. This paper introduces a selfdeveloped framework that integrates multiple groups of biological datasets and hierarchical network information, which aims to improve the accuracy of pathogenic gene prediction.

1.2 Gene Behavior as a Molecular Phenomenon

Gene behavior can be understood as a molecular-level process governed by genetic variations, regulatory mechanisms, and interactions within biological systems. Variations in genes, such as mutations or alterations in genomic sequences, can significantly influence cellular functions and may lead to abnormal biological outcomes. These changes disrupt the normal balance of cellular processes, including gene expression, regulation, and signaling pathways. Unlike external factors, such variations originate within the genome and accumulate over time, contributing to functional differences in gene activity [5], [6], [7].

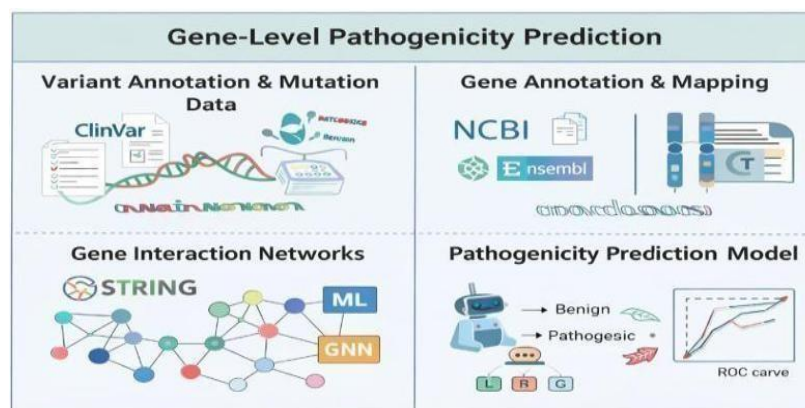


Figure 1.1 Gene - Level Prediction System

Figure 1.1 illustrates the proposed gene-level pathogenicity prediction system, showing how multiple biological datasets such as ClinVar, NCBI, and STRING are integrated. It highlights the use of machine learning and graph

neural networks to classify genes as pathogenic or benign by combining feature-based and network- based information.

Mutations in genes often affect their functional roles by altering the encoded biological processes or disrupting regulatory mechanisms. Genes associated with critical cellular functions, such as DNA repair, signaling pathways, and metabolic regulation, are particularly sensitive to such variations. In many cases, the impact of these mutations becomes evident only when considered in the broader context of gene interactions and network-level dependencies rather than in isolation [4].

Gene functionality is closely linked to its interaction with other genes and proteins within complex biological networks. These interactions determine how biological signals are transmitted and how cellular processes are coordinated. Even minor variations can influence interaction patterns, leading to changes in functional outcomes.

Understanding gene-level behavior necessitates a comprehensive approach that goes beyond traditional single-dataset analysis. By integrating variant data, gene annotations, and interaction networks, it becomes possible to capture both local and global biological context. This multi-level perspective enables more accurate identification of gene characteristics and supports the development of predictive models for classifying genes as pathogenic or benign [5], [6], [7].

1.3 Role of Gene Interactions in Biological Systems

Biological systems have a core feature of high interconnectivity: genes and proteins interact through complex networks. The academic community widely uses protein–protein interaction (Protein–Protein Interaction, PPI) network modeling, which takes genes as nodes and interactions as edges, to clarify the mechanisms of mutual influence between genes within cells.

Gene interaction is a core mechanism that regulates various biological activities and maintains cellular homeostasis. A single gene mutation can affect multiple genes through shared pathways, and interpretations derived from the isolated analysis of individual single genes are often one-sided. The interaction networks of these genes can identify key high-connectivity genes, and integrating these networks into computational models can improve prediction performance and deepen biological insights.

1.4 Limitations of Traditional Approaches

Traditional gene analysis takes sequence analysis and statistical methods as its core foundations. While this approach can provide useful biological information, it fails to capture the interconnected nature of biological systems. Most models used in this field also treat genes as independent entities, overlooking network-level associations.

Traditional methods in the field of computational genomics suffer from two core flaws. First, they rely on single-source datasets, and the impact of biological information heterogeneity often leads to an incomplete understanding of gene behavior. Second, they cannot efficiently process large-scale data or model complex nonlinear relationships, which drags down prediction accuracy and limits their applicability. As such, there is an urgent need to introduce advanced artificial intelligence (AI) methods to overcome these challenges.

1.5 Need for Data Integration

Biological heterogeneous data originates from four categories: genomic databases, mutation knowledge bases, interaction networks, and population studies. Any single data source can only provide partial information about gene behaviour, so integrating multiple data sources is a necessary prerequisite for gaining a comprehensive understanding of biological systems.

Data integration in the bioinformatics field refers to the process of combining complementary information—including variant annotations, gene mappings, mutation frequencies, and biological interaction relationships—to form a unified representation. This approach can optimize feature extraction and improve modeling quality. However, format differences across biological data sources create notable challenges, so preprocessing and mapping technologies are required to guarantee data accuracy, which in turn supports models to capture the complexity of biological systems.

1.6 Role of Machine Learning in Gene Prediction

At present, Machine Learning has become an important tool for large-scale genomic data analysis. It can learn latent patterns from biological data, and complete gene classification tasks relying on mutation counts, gene annotations, and functional characteristics.

In the gene prediction task, the four traditional machine learning algorithms of random forest, support vector machine, logistic regression, and XGBoost, despite being able to process high-dimensional biological data and capture associations between features, are only compatible with tabular data, fail to account for gene-gene interactions, and cannot capture the critical biological dependencies within gene networks.

1.7 Graph Based Learning in Bio-Informatics

Graph-based learning provides a powerful general analytical framework for representing various complex biological systems. In our study, we map the genes included in the research to nodes and biological interactions to edges; this representation enables computational models to synchronously analyze two types of information: feature-level and network-level data.

The core innovation of graph neural networks is to incorporate graph topological structures into the learning process, which extends the capability boundaries of traditional machine learning. Mainstream models include GCN, GAT and GraphSAGE; these models can capture the complex interactions in biological systems, capture both

local and global correlations of genes, and improve the accuracy of pathogenicity prediction as well as the biological interpretability of the model.

1.8 Proposed Approach

This paper proposes to build an integrated AI-driven gene-level pathogenicity prediction system, which uses machine learning and graph neural network technologies to preprocess different types of authoritative biological datasets including ClinVar, NCBI Gene, STRING interaction network and population mutation datasets.

The bioinformatics modeling scheme proposed in this paper first completes the cleaning, filtering, and mapping of raw biological data via a multi-stage preprocessing pipeline to generate structured gene representations, then constructs two types of features in parallel that are respectively adapted for traditional machine learning and graph neural networks; after feature fusion, the scheme not only improves prediction accuracy, but also enhances interpretability.

1.9 Significance of the Work

This study proposes an efficient analytical framework for pathogenic gene prediction, which integrates multiple types of biological data and interaction networks, combines feature learning and graph learning algorithms, and outperforms previous conventional research methods.

Introducing machine learning (Machine Learning) and graph neural networks (Graph Neural Networks) into biomedical research can improve the accuracy of gene behavior prediction, and also support future exploration in fields including computational genomics and disease prediction.

2. Literature Review and Problem Formulation

Preamble to the Chapter

The rapidly expanding collection of genomic and various types of biological data has strongly driven the development of the fields of computational genomics and bioinformatics. Gene pathogenicity prediction can support disease diagnosis, biomedical analysis, and genomic research, yet it is difficult to achieve accurate prediction due to the complexity of biological systems and the interconnected nature of genes.

In the current genome analysis field, traditional research methods suffer from a host of inherent limitations. The emerging computing technologies we have introduced can effectively resolve these industry-wide pain points. This chapter will proceed sequentially to elaborate on five core tasks, the first of which is sorting out the research background of this field.

2.1 Understanding Theoretical Background and Theoretical Basis

2.1.1 Structure and Function of Genes

Genes are the basic biological units that store organisms' genetic information and regulate cellular life processes. Composed of nucleotide sequences, their core structure consists of three core components: coding regions, regulatory regions, and functional elements that affect biological activity.

Gene mutations can disrupt the normal operation of genes, interfere with core biological processes inside cells, and subsequently trigger a wide range of abnormal biological states. Understanding the structure and function of genes is a necessary prerequisite for analyzing their behavior and identifying pathogenic characteristics. Genes operate within interconnected biological systems, and their functions are influenced by interactions with other genes and proteins. Independent analyses therefore often cannot provide a complete biological interpretation. [5],[6].

Gene function is not solely determined by its sequence but also by its interactions within biological systems. Genes operate within complex networks where they interact with other genes and regulatory elements to control various cellular processes. These interactions influence gene activity, signaling pathways, and biological responses. Changes in gene structure or expression can disrupt these interactions, leading to altered functional outcomes [4].

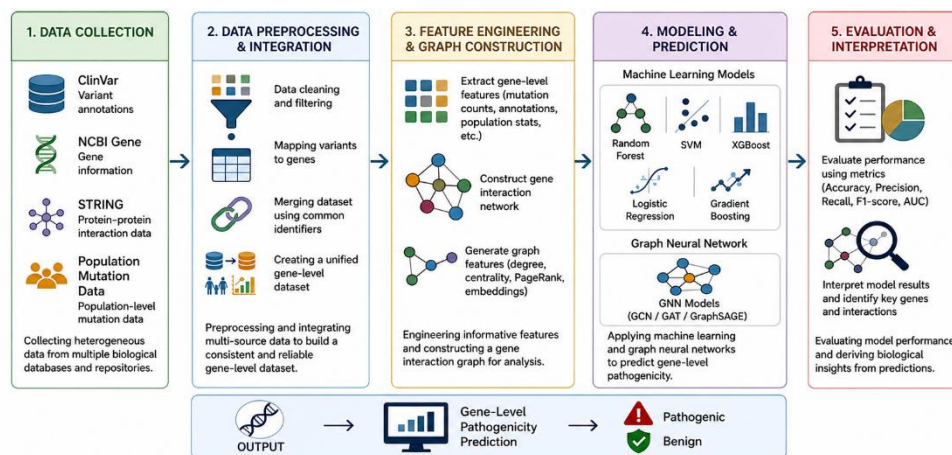


Figure 2.1 Computational approach in Gene - Level Pathogenicity Prediction

Figure 2.1 presents the computational workflow for gene-level pathogenicity prediction, illustrating how multi-source biological data is collected, preprocessed, and integrated to extract gene-level features. It demonstrates the use of machine learning and graph neural network models to analyze both feature-based and interaction-based.

Understanding both the structural and functional aspects of genes is essential for analyzing gene behavior. By considering gene-level features along with interaction-based relationships, it becomes possible to develop computational models that more accurately represent biological systems. This forms the theoretical basis for gene-level pathogenicity prediction in the proposed work [2],[3].

2.1.2 Application of Computational Genomics in Gene-Level Analysis

This paper defines computational genomics as an interdisciplinary field that combines biology and advanced computing technologies, and outlines four categories of public databases, including ClinVar, that provide core information such as genetic variants. Traditional experimental gene analysis suffers from the drawbacks of long processing times and high costs. Modern computational genomics can efficiently process largescale genomic datasets, and leverages machine learning and graph-based learning to process heterogeneous biological data, supporting scalable biomedical research.

2.1.3 Role of Machine Learning in Gene Prediction

At present, machine learning has been widely applied to genomic data analysis and pathogenicity prediction. It can learn from manually constructed gene-level features, and complete gene classification by relying on mutations, annotations, and biological properties.

Currently, the four machine learning algorithms commonly used for gene prediction tasks are random forest, logistic regression, SVM, and XGBoost. These algorithms can process high-dimensional biological data and capture the complex feature associations between various types of variables.

2.1.4 Role of Graph Neural Networks in Biological Systems

This study introduces graph neural networks as an effective framework for modeling biological interaction networks, with genes set as graph nodes and biological interactions set as the graph's edges.

Unlike traditional machine learning, graph neural networks (GNNs) integrate graph topological structures into their learning processes. Mainstream GNN models include GCN, GAT, and GraphSAGE. These models can capture both local and global correlations in biological systems, which not only improves prediction accuracy but also deepens understanding of biological mechanisms. [2],[3],[4].

2.2 Review of Existing Literature Survey

2.2.1 Genomic and Omics-Based Gene Analysis

Numerous studies in the field rely on large-scale biological datasets, using genomic and omics analyses to investigate gene behaviors and mutational characteristics, and cover three core research directions. Although these studies hold research value; they have notable limitations: they depend on isolated data and lack interaction analysis. [5],[6],[7].

However, these methods mainly rely on sequence-level or mutation-level data and often treat genes as independent entities. They do not effectively capture the interaction- based relationships between genes, which play a crucial role in biological processes. As a result, they may not fully represent the functional behavior of genes [4].

2.2.2 Feature-Based Gene Analysis Using Classical Methods

Feature-based methods in the field of biological prediction conduct analysis relying on manually designed biological features. These methods commonly use algorithms such as decision trees, random forests, and SVMs. While their prediction accuracy outperforms that of purely statistical methods, they have an inherent flaw: they ignore the dependencies among biological interactions.

Multiple recently published studies have applied advanced machine learning paired with structured feature representation to gene-level prediction. Techniques including feature engineering, dimensionality reduction to improve efficiency, ensemble learning, and boosting all deliver excellent performance in genomic classification. [8],[9],[10].

However, these approaches primarily focus on feature-based representations and do not capture the interaction-based relationships between genes. Biological systems are highly interconnected, and ignoring these dependencies limits the ability to model complex gene behavior. Therefore, while machine learning models provide strong predictive

capabilities, they need to be complemented with graph-based approaches to fully capture biological context and improve prediction accuracy [2],[3].

2.2.3 Feature-Based Representations and Machine Learning Models

Multiple recently published studies have applied advanced machine learning paired with structured feature representation to gene-level prediction. Techniques including feature engineering, dimensionality reduction to improve efficiency, ensemble learning, and boosting all deliver excellent performance in genomic classification. [8],[9],[10].

Table 2.1 Genomic and Computational Approaches for Gene-Level Prediction

Model	Category Definition	Core Operations	Application Directions	Core Limitations
Variant-level analysis	Annotates single-base pathogenic variants relying on the ClinVar dataset	Basic variant screening	Used for basic variant screening	It cannot integrate the synergistic effects of multiple variants
Gene-level feature analysis	Aggregates intragenic variant scores	Suitable for candidate gene screening	Candidate gene screening	It loses detailed interaction information between variants
Machine learning models	Equipped with random forests, SVM, and XGBoost	Fit the associations between variants & phenotypes	Applicable to large-sample risk prediction	They cannot capture network-level associations
Graph neural networks	Model molecular interaction networks	Complex pathway analysis	Suitable for complex pathway	They have a high dependence on training data

Hybrid ML+GNN model	Integrates the advantages of the two types of models	Integrates the advantages of the two types of models	Cutting-edge hybrid model	Insufficient interpretability
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Table 2.1 summarizes genomic and computational approaches for gene-level prediction, highlighting their methodologies, applications, and key limitations. It provides a comparative overview of traditional feature-based, network-based, and hybrid learning techniques used for gene classification.

2.2.4 Sequence-Based Gene Analysis

In the field of genomic research, the core of mainstream sequence-based pathogenic gene analysis methods is to identify pathogenic genes by relying on nucleotide sequence patterns and mutation information. Traditional sequence alignment and statistical techniques are widely applied in these methods, but they cannot capture biological associations at the network level, and thus have limited predictive performance for complex biological systems.[5]

2.2.5 Ensemble Learning in Gene-Level Prediction

Multiple recently published studies have applied advanced machine learning paired with structured feature representation to gene-level prediction. Techniques including feature engineering, dimensionality reduction to improve efficiency, ensemble learning, and boosting all deliver excellent performance in genomic classification. [2],[3].

2.3 Critique of Existing Techniques

Current gene prediction technologies suffer from numerous core limitations: traditional statistical and sequence methods cannot capture biological interactions, most machine learning models can only process tabular features and ignore graph dependencies, single-source datasets limit models’ generalization ability and are unable to process

heterogeneous large-scale genomic data, and these gaps have driven the development of integrated AI-driven frameworks.[8],[10].

2.4 Identification of Research Gaps

From the review of existing literature, several research gaps can be identified in gene-level prediction. Most current approaches rely heavily on feature-based models and do not effectively incorporate interaction-based relationships between genes. As a result, they fail to capture the complex dependencies present in biological networks, leading to limited prediction accuracy [4]. Another important gap is the lack of effective integration of heterogeneous biological datasets. While multiple data sources such as variant data, gene annotations, and interaction networks are available, many existing methods do not utilize them together in a unified framework. This results in incomplete representation of gene behavior [5],[6],[7].

Table 2.2 Literature Review

No.	Author (Year)	Key Methodology / Findings	Limitation / Gap
1	H. Chen et al. (2018)	Applied deep learning techniques in drug discovery and biomedical prediction tasks.	Focused mainly on drug discovery, not gene-level pathogenicity prediction.
2	T. N. Kipf and M. Welling (2017)	Introduced Graph Convolutional Networks (GCNs) for semi-supervised graph classification.	Limited scalability for very large biological graphs.
3	W. L. Hamilton et al. (2017)	Proposed GraphSAGE for inductive representation learning on large graphs.	Did not specifically address biomedical gene prediction tasks.
4	P. Veličković et al. (2018)	Developed Graph Attention Networks (GAT) using attention mechanisms in graphs.	Computational complexity increases with graph size.

5	D. Szklarczyk et al. (2021)	STRING database provided large-scale protein–protein interaction networks.	Interaction data may contain noisy or incomplete relationships.
6	R. Landrum et al. (2018)	ClinVar improved access to clinically annotated genetic variants.	Variant annotations are incomplete for many genes.
7	NCBI Resource Coordinators (2022)	Provided comprehensive biological and genomic databases.	Requires integration from multiple heterogeneous sources.
8	1000 Genomes Project Consortium (2015)	Created a global reference for human genetic variation.	Population-level data lacks functional pathogenic interpretation.
9	L. Breiman (2001)	Introduced Random Forest for robust ensemble classification.	Performance depends heavily on feature engineering.
10	C. Cortes and V. Vapnik (1995)	Proposed Support Vector Machines for classification tasks.	Struggles with very high-dimensional biological datasets.
11	T. Chen and C. Guestrin (2016)	Developed XGBoost for scalable gradient boosting.	Requires hyperparameter tuning for optimal performance.
12	A. Grover and J. Leskovec (2016)	Introduced node2vec for graph feature learning.	Does not directly utilize node attributes effectively.
13	F. Pedregosa et al. (2011)	Developed Scikit-learn framework for machine learning	General-purpose library without biomedical specialization.
14	D. P. Kingma	Proposed Adam optimizer for	Sensitive to learning rate

	and J. Ba (2015)	efficient deep learning training.	selection in some cases.
15	T. Hastie et al. (2009)	Discussed statistical learning methods for predictive modeling.	Traditional methods lack graph-based biological context.
16	S. Raschka and V. Mirjalili (2019)	Presented practical machine learning techniques using Python.	Primarily educational without domain-specific optimization.
17	J. Leskovec and A. Krevl (2014)	SNAP datasets enabled analysis of large-scale graph networks.	Generic graph datasets may not fully represent biological .
18	T. Mikolov et al. (2013)	Proposed efficient representation learning using embedding techniques.	Designed mainly for language modeling, not biological networks.
19	F. Chollet (2018)	Demonstrated deep learning applications using Python frameworks.	Focused more on implementation than biomedical interpretation.
20	J. Brownlee (2016)	Explained practical machine learning workflows and model .	Lacked integration with graph neural network methodologies.

Table 2.4 provides a concise literature review of recent studies related to gene-level prediction using computational, machine learning, and graph-based methods. It also highlights the key research gaps addressed in the proposed work, including data integration, interaction modeling, and the need for hybrid ML and GNN approaches.

2.5 Problem Formulation

In the field of gene pathogenicity prediction, accurately distinguishing between pathogenic and non-pathogenic genes remains a core challenge. This problem has three major underlying causes. At present, traditional computational methods cannot capture biological associations at the feature and network levels, so the field urgently requires an

integrated framework equipped with four core capabilities. This study will develop this framework to support efficient and scalable gene pathogenicity prediction.

2.6 Chapter Summary

This chapter first lays out the core theoretical background of gene behavior, computational genomics, machine learning, and graph neural networks. It then reviews existing literature in the field of gene-level prediction, analyzes the limitations of traditional methods, develops an integrated pathogenicity prediction framework built on identified research gaps, and clarifies the core research questions of this work. [1],[2],[3]

3. Problem Statement

Preamble to the Chapter

The core goal of this chapter is to formally put forward the central question of this study on gene-level pathogenicity prediction, and systematically explain the importance, core objectives, and expected outcomes of the research. Unlike the previous chapter, whose only positioning was to sort out gaps and limitations in existing literature,

This chapter converts the aforementioned literature gaps into a clear problem statement and delineates the specific scope for carrying out the research. We require the computational models that support this study to have both high prediction performance and the ability to capture the inherent associations in gene-level data. This chapter will integrate

multi-source biological datasets, combine machine learning and graph methods to model gene interactions, and list the assumptions and limitations of this study to ensure transparency, so as to lay a solid foundation for the hybrid pathogenicity prediction framework proposed in subsequent work.

3.1 Background of the Problem

The behavior of genes is by no means an independent trait of individual units, but a complex process co-regulated by multiple types of molecular mechanisms and biological pathways. Although conventional single-dimensional genomic and variant analysis can effectively characterize basic mutational patterns, it cannot fully capture the functional connections between genes within biological systems: all genes exist in interconnected networks, and their behavior is influenced not

Only by variations in their own sequences, but also by interactions with other genes and various regulatory processes . Isolated single-dimensional analysis is completely unable to reveal these layer-by-layer nested interdependencies, and only by relying on gene-gene interaction (GGI) networks can the functional impacts of genetic variants be reliably studied [4].

In early years, the extreme scarcity of integrable biological datasets and computational methods once restricted large-scale analysis of gene interactions. Today, public resources including ClinVar, NCBI Gene, and STRING have become widely accessible, and are sufficient to support the integration of multi-source data and the development of predictive models.[5],[6],[7].

3.2 Problem Statement

Gene-level pathogenicity prediction is a complex and highly challenging computational problem. Constrained by the heterogeneity of biological datasets and gene interconnectivity, traditional statistical sequence methods fail to effectively capture interactions and only deliver limited accuracy. Existing machine learning approaches only focus on features, while neglecting the core value of biological networks. [4]

Therefore, the problem addressed in this work is the development of an integrated computational framework capable of:

- Combining multiple biological datasets
- Extracting meaningful gene-level features
- Modeling gene interaction networks
- Applying Machine Learning and Graph Neural Networks
- Improving prediction accuracy and biological interpretation

The proposed framework aims to classify genes as pathogenic or non-pathogenic using both feature-level and interaction-level information.[8],[10].

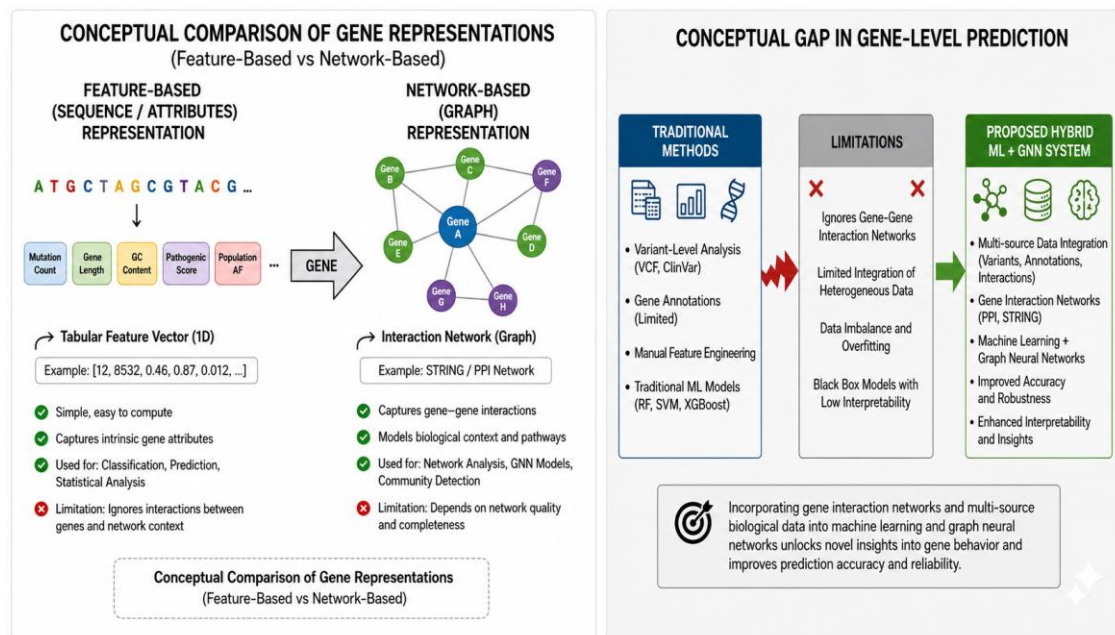


Figure 3.1 Conceptual Gap in Gene-Level Prediction

Figure 3.1 Highlights the conceptual gaps in existing gene-level prediction approaches, emphasizing the lack of effective integration of gene interaction networks and the limited use of interpretable and hybrid learning models in current methodologies.

3.3 Aim of the Project

This project takes identifying pathogenic genes and constructing disease risk models as its core tasks. It designs a hierarchical multi-scale prediction framework that integrates variant-level genomic evidence, gene-level mutation burden, and topological features of protein-protein interaction networks. By combining traditional machine learning and graph neural networks, this project aims to improve prediction performance, enhance biological interpretability, and support reliable disease risk assessment at both the gene and patient levels.

3.4 Objectives of the Project

For this purpose, the following objectives are formulated:

1. To collect and preprocess raw genomic datasets and transform variant-level mutation data into aggregated gene-level features.
2. To construct a gene interaction graph using STRING Protein–Protein Interaction data and integrate network information with gene-level features.
3. To implement machine learning models and Graph Neural Network architectures for pathogenic gene prediction.
4. To evaluate model performance through comparative analysis and analyze the impact of biological network information on prediction accuracy.

Table 3.1 Project Objectives and Methods

Obj	Description	Methodology
OB1	Dataset collection, Preprocessing	Multi-source integration (ClinVar, NCBI Gene, STRING, 1000 Genomes), Cleaning, variant-to-gene mapping, and feature engineering
OB1	Data Mapping	Construct a gene interaction graph using PPI data and integrate network information with gene-level features.
OB3	Model training	Hybrid Machine Learning (RF, SVM, LR+ GB , XGBoost) + GNN (GCN, GraphSAGE).
OB4	Evaluation & Deployment	Validation using accuracy, precision, recall, and confusion matrices, streamlit web-based interface.

Table 3.1 summarizes the major objectives of the proposed work along with the methodologies and computational tools used for implementation. The table provides an overview of the different stages involved in the development of the gene-level pathogenicity prediction framework.

3.5 Summary Of The Proposed Methodology

The bio-omics modeling methodology proposed in this study encompasses six core research stages. In the initial phase of the research, data is collected from three major genomic databases including ClinVar. Three types of outliers are cleaned, gene mapping is completed, and the entire workflow is clearly annotated, which fully guarantees the reproducibility of the research methods. [5],[6],[7],[4]

This study establishes a complete machine learning modeling pipeline tailored for bioinformatics scenarios. We first complete data preprocessing: leveraging feature engineering, we extract three categories of biological attributes—mutation frequency, interaction [8][10] counts, and annotation-based features—to construct a gene interaction graph where nodes represent genes and edges represent biological interactions. We then deploy two types of models, and ultimately complete performance evaluation using six classification metrics including Accuracy.

3.6 Assumptions in the Study

The proposed work is based on the following assumptions:

- The collected biological datasets contain reliable and valid genomic information.
- Gene interaction networks accurately represent biological relationships.
- Preprocessing techniques effectively remove inconsistencies from datasets.
- Machine Learning and Graph Neural Network models can learn meaningful biological patterns from data.
- The available computational resources are sufficient for model training and evaluation.

3.7 Constraints and Limitations

The proposed framework has certain limitations and constraints:

- Biological datasets may contain incomplete or noisy information.
- Graph Neural Networks require high computational resources for training.
- Large-scale genomic networks increase computational complexity.

- Biological interpretations depend on the quality of interaction data.
- Limited availability of labeled pathogenic gene datasets may affect model generalization.

Table 3.2 Summary of the constraints and limitations on the proposed system

Category	Specific Constraint	Impact on Performance / Deployment
Data Dependency	Reliance on multi-source datasets (ClinVar, NCBI Gene, STRING).	Accuracy is tethered to data quality and correct cross-dataset mapping.
Computational Cost	Graph construction and GNN training on large networks.	Increased memory usage and processing time for gene networks.
Inference Latency	Hybrid ML + GNN framework using ensemble techniques.	Combining multiple models increases the time required for a single prediction.
Fixed Input Size	Dependence on engineered gene-level features.	Incomplete or poorly designed features can significantly reduce model effectiveness.
Biological Scope	Dependence on accuracy of interaction networks (PPI/STRING).	Noisy or missing interaction data negatively impacts overall prediction reliability.
Interpretability	Complex hybrid architecture (ML + GNN).	While results are more accurate, explaining the specific logic behind a "Pathogenic" decision remains difficult.

***Table 3.2** outlines the key constraints and limitations associated with the proposed gene-level prediction system, highlighting challenges related to data integration, feature representation, graph quality, and computational complexity. It clarifies the scope and practical boundaries of the study.*

3.8 Expected Outcomes

The expected outcomes of the project are as follows:

- Development of an integrated computational framework for pathogenic gene prediction.
- Improved classification accuracy using Machine Learning and Graph Neural Networks.
- Better representation of biological interactions through graph-based learning.
- Enhanced biological interpretability and understanding of gene behavior.
- Scalable framework capable of processing large-scale genomic datasets.

3.9 Significance of the Proposed Work

The authors of this paper propose a novel research framework in the field of computational genomics. This framework integrates multiple types of biological datasets, and combines feature-based learning and graph interaction modeling. Compared with traditional single-dimensional analysis methods, it can simultaneously capture the local features and global dependencies of genes, improve prediction accuracy, and support biomedical applications such as disease prediction. [17][13][7]

3.10 Chapter Summary

This chapter covers core research modules including the problem statement, research objectives, and other key foundational research components, and presents supporting content such as research hypotheses, constraints, and research significance. The core gene-level pathogenicity prediction framework integrates multiple types of components, and is designed to improve prediction accuracy and interpretability.

4. Problem Solving

Preamble to the Chapter

This chapter introduces the detailed research approach, data processing pipeline, and model development strategy of the gene-level prediction system we propose. It breaks down the full end-to-end workflow that transforms multi-source biological data into an integrated interaction network, and elaborates on all system components from computational, biological, and analytical perspectives, to ensure the research is clearly reproducible, scientifically sound, and effective.

This study integrates the two types of learning paradigms, feature-based learning and interaction-based learning, and combines gene-level attributes with gene interaction networks. It first converts variant-level mutation data into gene-level representations and constructs a gene association graph structure, then uses a hybrid framework composed of machine learning models and Graph Neural Networks (GNNs) to simultaneously learn the independent features of genes and their network dependency relationships.

4.1 Overall System Architecture

This paper proposes a modular end-to-end system architecture specifically designed for gene-level prediction. Its core execution process is:

- sequentially divided into: multi-source data
- Collection and integration, preprocessing and feature mapping, gene interaction
- Network construction, machine learning-based feature learning,
- Graph neural network-based network learning, prediction
- Visualization, and user interface deployment.

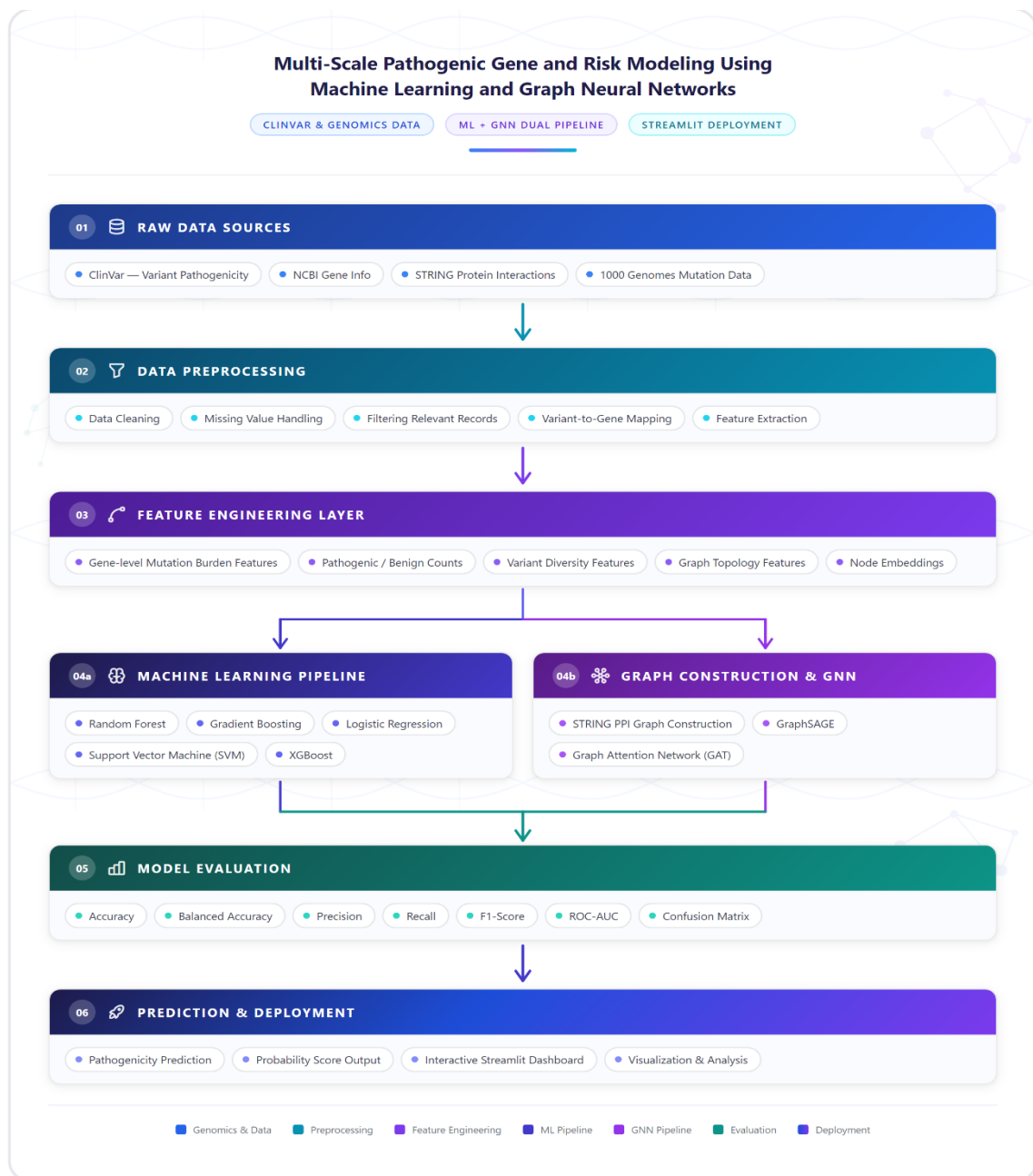


Figure 4.1 System Architecture Diagram

Figure 4.1 illustrates

- The workflow begins with raw biological, mutation, and interaction data.
- Data preprocessing prepares clean and structured gene-level information.
- Data mapping aligns mutation records, gene identifiers, and network relations into one common framework.

- The final processed dataset is used in two parallel branches: ML-based prediction and graph-based prediction.
- The ML branch uses tabular biological features for classification.
- The graph branch constructs a STRING PPI network and applies GNN models for relational learning.
- The models are evaluated using statistical metrics and visualization techniques.
- The best-performing framework is deployed through a Streamlit dashboard for user

4.2 Dataset Details and Curation

4.2.1 ClinVar Database

- **Database Used:** ClinVar
- **NCBI Taxonomy ID:** 9606
- **Attributes Extracted:**
 - Variant ID
 - Gene Symbol
 - Clinical Significance (Pathogenic / Benign)
 - Mutation Type
 - Genomic Location

VariationID	GeneSymbol	Chromosome	Start	ReferenceAllele	AlternateAllele	ClinicalSignificance	Type
0	3	AP5Z1	7	4787730	na	na	Pathogenic
1	5	FOXRED1	11	126275389	na	na	Pathogenic
2	6	FOXRED1	11	126277517	na	na	Likely pathogenic
3	14	HFE	6	26093008	na	na	Benign
4	18	HFE	6	26093215	na	na	Pathogenic
5	20	WDR35	2	19989284	na	na	Pathogenic

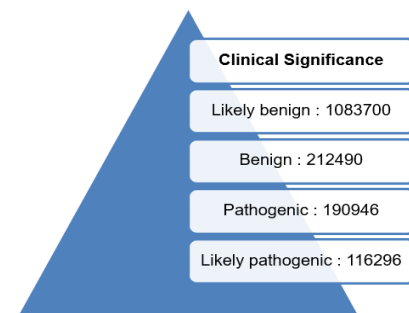
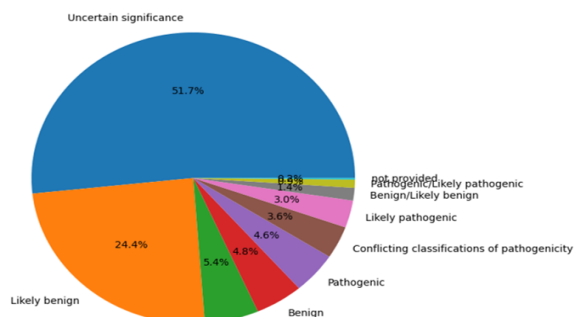


Figure 4.2 Dataset Structure & pie chart

Figure 4.2 Show the dataset details in terms of Attributes and Statistics

4.2.2 NCBI Gene Database (Gene Mapping)

- **Database Used:** NCBI Gene
- **NCBI Taxonomy ID :** 9606

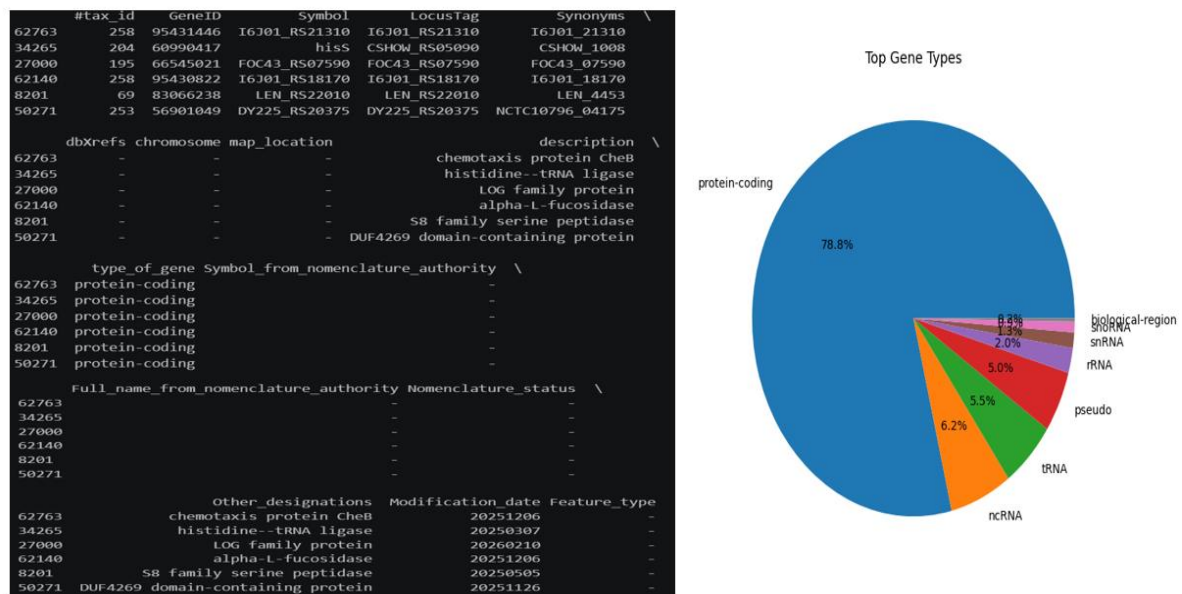


Figure 4.3 Dataset Structure & pie charts

Figure 4.3 Show the dataset details in terms of Attributes and Statistics

4.2.3 STRING Database (Interaction Network)

- **Database Used:** STRING (Search Tool for the Retrieval of Interacting Genes/Proteins)

4.2.4 Population Mutation Dataset (1000 Genomes Project)

- **Database Used:** 1000 Genomes Project
- Coverage: Global human population samples
- Dataset Statistics
- **Total Genes:** ~20,000+

- **Total Variants Processed:** ~1,000,000+
- **Class Distribution:**
 - **Pathogenic Class:** Minority class
 - **Benign Class:** Majority class

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	\
0	1	10177	.	A	AC	100	PASS
1	1	10235	.	T	TA	100	PASS
2	1	10352	.	T	TA	100	PASS
3	1	10505	.	A	T	100	PASS
4	1	10506	.	C	G	100	PASS

	INFO	FORMAT	HG00096
0	AC=2130;AF=0.425319;AN=5008;NS=2504;DP=103152;...	GT	1 0
1	AC=6;AF=0.00119808;AN=5008;NS=2504;DP=78015;EA...	GT	0 0
2	AC=2191;AF=0.4375;AN=5008;NS=2504;DP=88915;EAS...	GT	1 0
3	AC=1;AF=0.000199681;AN=5008;NS=2504;DP=9632;EA...	GT	0 0
4	AC=1;AF=0.000199681;AN=5008;NS=2504;DP=9676;EA...	GT	0 0

Figure 4.4 Dataset Attributes

Figure 4.4 Show the dataset details in terms of Attributes

4.3 Dataset Curation

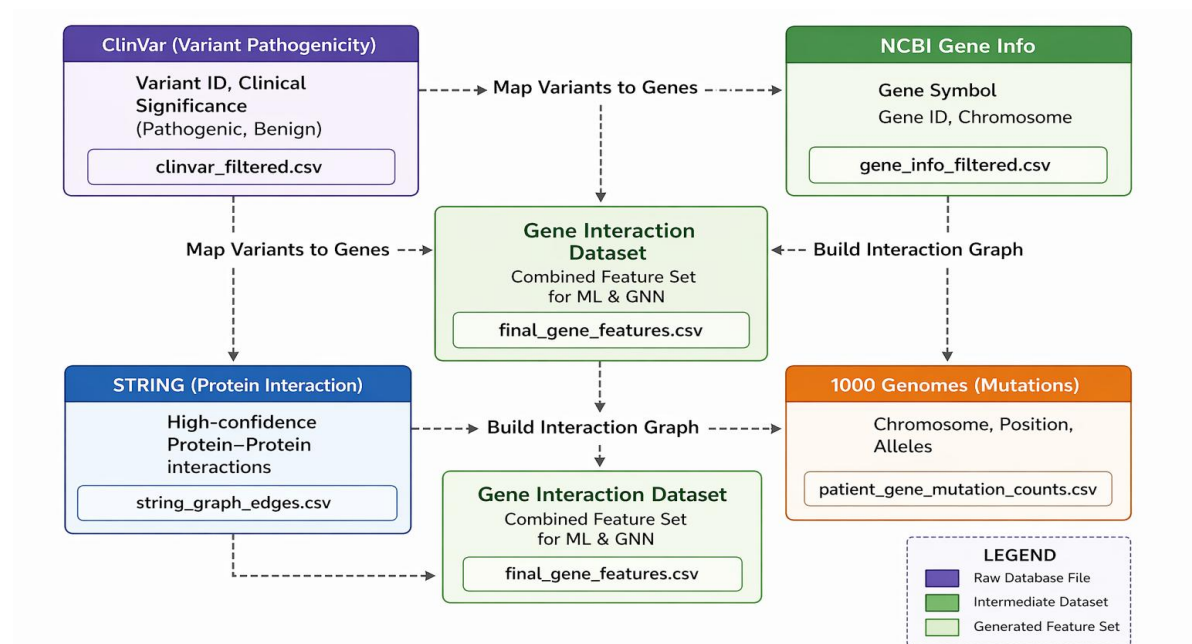


Figure 4.5 Dataset Curation

Figure4.5 *The proposed system integrates multiple heterogeneous genomic datasets to construct a unified gene-level feature representation for pathogenicity prediction. The overall mapping process is illustrated*

4.3.1 Explanation of Dataset Curation

1. ClinVar provides variant pathogenicity information, which is mapped to corresponding genes using chromosomal positions.
2. NCBI Gene standardizes the GeneSymbol and GeneID across the complete dataset.
3. STRING provides high-confidence protein-protein interaction data to construct a gene interaction map.
4. 1000 Genomes is used to extract mutation information matched with ClinVar variants to calculate the number of patient mutations.
5. Variant data is aggregated to form gene-level mutation burden features.
6. All datasets are integrated using universal keys.
7. The final output file, finalgenefeatures.csv, is generated for use by machine learning (ML) and graph neural network (GNN) models.

4.4 Pre-Processing of Gene Data for Modeling

- In the biological data preprocessing stage of this study, the core task is to convert raw biological data into a format compatible with machine learning and graph-based models. We then complete data cleaning work including deduplication, missing value processing, and format standardization for three types of data: variation data, gene annotation data, and interaction network data.
- This study is based on the NCBI Gene database. First, we map omics variation data to gene features, extract two core associated attributes—mutation count and annotation, and define a gene interaction graph that takes genes as nodes and interactions as edges.
- The final pre-processed dataset of this study contains tabular and graph-structured features, which can support efficient learning for hybrid machine learning and graph neural networks.

4.5 Overview of the Pre-Processing Pipeline

The biological data preprocessing pipeline adopted in this study includes four core operations: cleaning and integrating multi-source biological data, mapping hierarchical variant information, constructing gene interaction networks, and standardizing cross-dataset data. This pipeline ensures data consistency and is suitable for similar biological omics research.

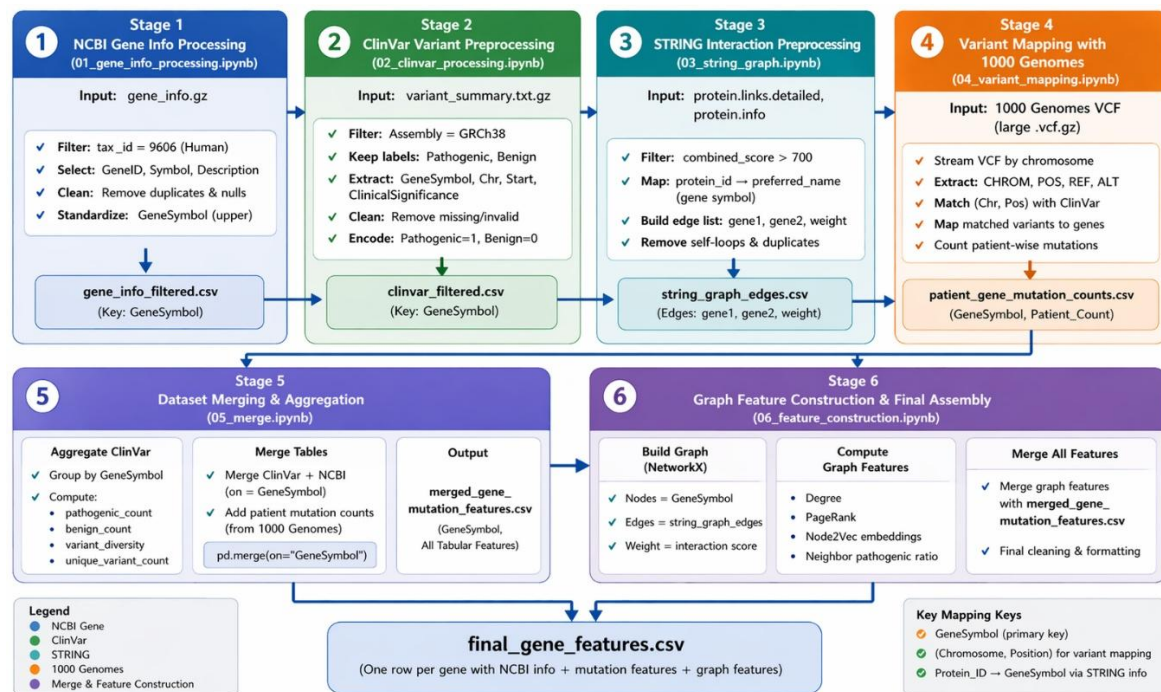


Figure 4.6 pre-processing

Figure 4.6 illustrates

- This preprocessing workflow processes NCBI human gene data, and extracts and filters three categories of relevant core gene information.
- Preprocess the ClinVar data: first screen for clinical variants, then extract gene mutation information.
- First, we map the population variant sites to genes, then calculate the number of mutations in each gene.
- First, merge multi-source datasets, then generate aggregated gene features.

- It is necessary to construct graph-based features and calculate three types of interaction indicators, namely degree, PageRank, and embedding.
- This study needs to integrate graph-based features and gene-level features to construct a unified, dedicated research dataset.
- Prepare a final dataset that can be directly applied to modeling work for machine learning and GNN models.

4.6 After pre-process results

4.6.1 Feature Distributions

The dataset used for modeling in this study integrates three categories of features: statistical features, graph structure features, and embeddings generated by Node2Vec. The first two feature categories are highly skewed and highly sparse, with most of their values concentrated at zero, which correspond to rare yet biologically significant events. In contrast, the embedding features follow an approximately normal distribution, making them suitable for machine learning applications. Additionally, the target variable suffers from class imbalance, which requires careful handling during model training to avoid prediction bias.

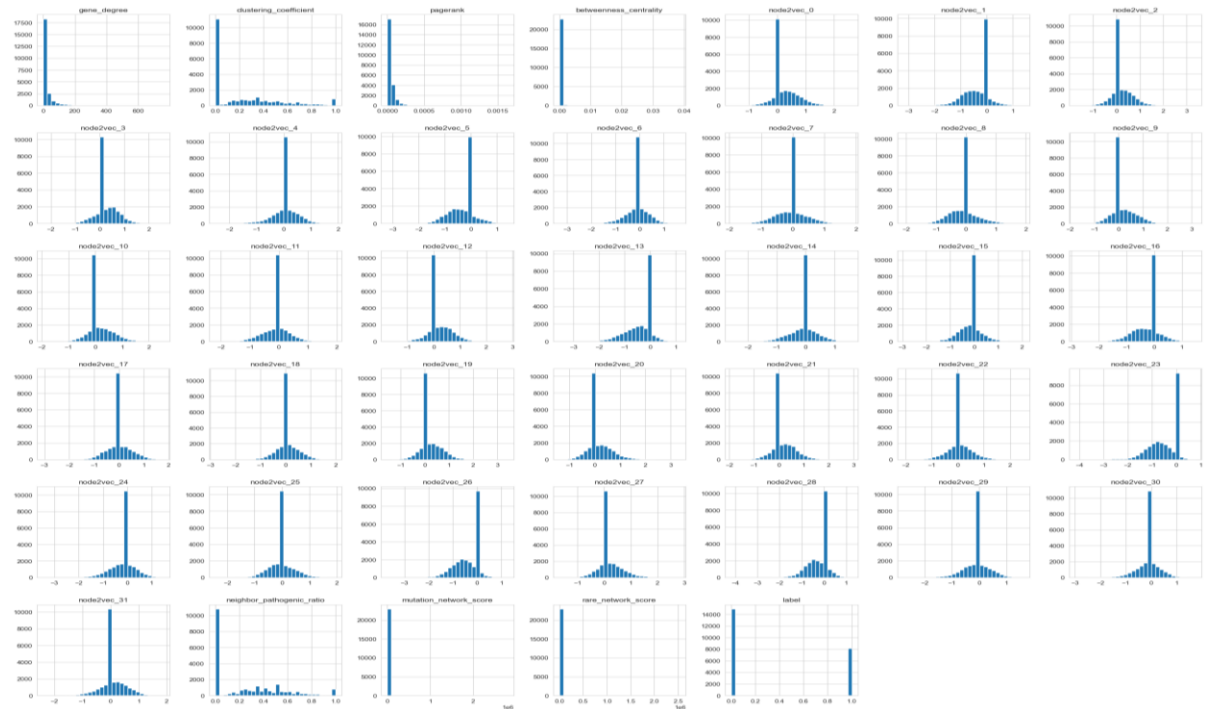


Figure 4.7 Feature Distributions of dataset Attributes

Figure 4.7 shows the distribution of genomic and graph-based features used for Pathogenic gene prediction. Most features exhibit skewed distributions with concentrated central values, indicating variability in gene connectivity, mutation patterns, and network characteristics across the dataset.

4.6.2 Gene Interaction Network (2D)

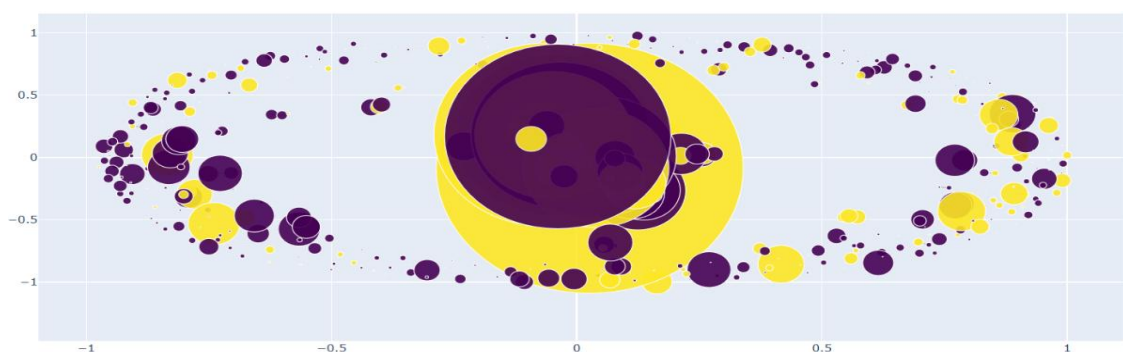


Figure 4.8 Gene interaction Network

Figure 4.8 illustrates

- Each node (circle) = Gene

- *Each edge (line) = interaction between genes*
- *Node size = importance (degree / connectivity)*
- *Node color = class (pathogenic vs benign)*
- *Large nodes = hub genes (highly connected)*
- *Small nodes = low-connectivity genes*
- *Presence of both classes → overlapping behavior*

4.7 Training Architecture

- This study constructs a two-branch machine learning and GNN pipeline for gene prediction tasks, with the entire workflow advancing in its actual execution order: First, data preprocessing is completed.
- The initial dataset contains 45 predictive features and 1 binary classification target label.
- We first remove uninformative columns such as GeneSymbol, description, leakage-prone attributes, and topology-related variables.
- Next, we use stratified sampling to split the dataset into an 80% training set and a 20% test set to preserve class balance.
- We also apply SMOTE with Tomek links to the training set to resolve class imbalance issues.
- For the machine learning branch, we select multiple classic models as base models, build a stacking ensemble centered on the tuned XGBoost, and optimize the decision threshold to maximize the F1 score.
- For the graph branch, we treat genes as nodes and STRING PPI interactions as edges, and adopt GraphSAGE and GATv2 to construct a graph neural network (GNN) to complete the modeling.

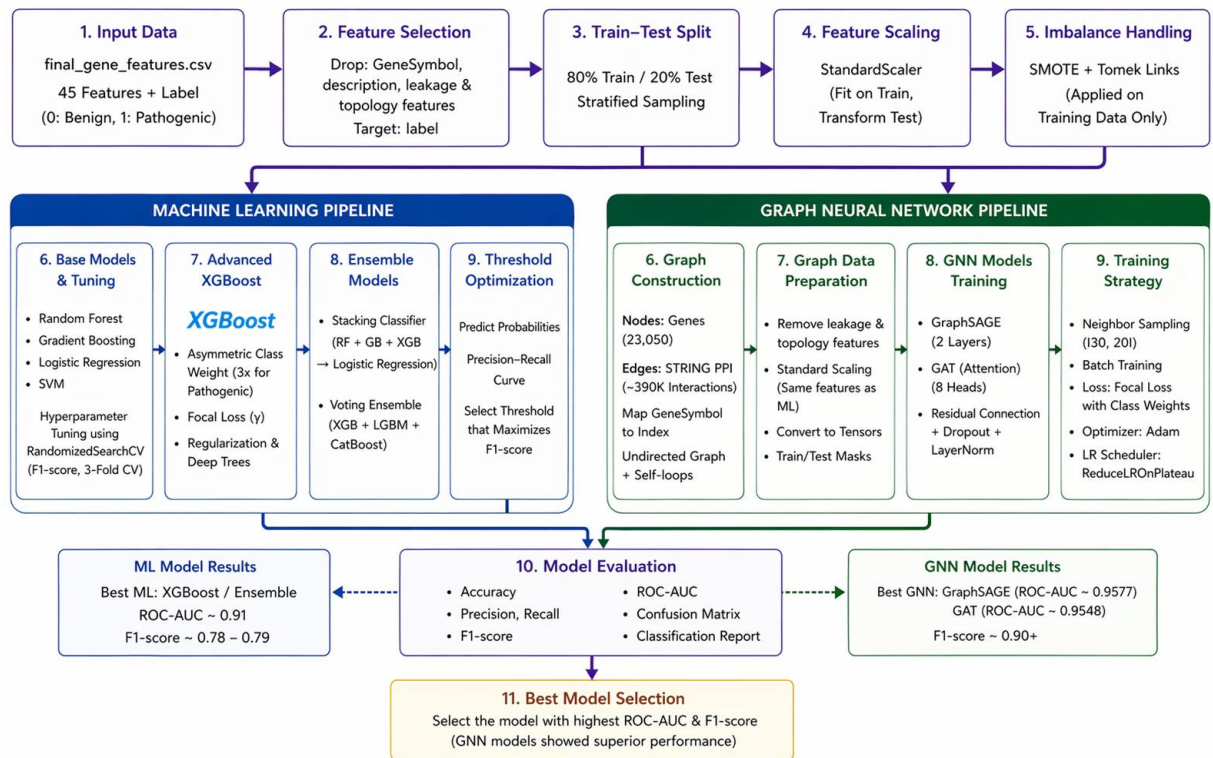


Figure 4.9 Training Architecture

Figure4.9 show complete workflow of the proposed prediction framework, integrating both ML & GNN pipelines

4.7.1 Data Parsing and Gene Mapping

Variant-level data is parsed and mapped to corresponding genes using gene identifiers. This process ensures that mutation information is aggregated at the gene level analysis.

```

1  gene_counts = {}
2
3  with open(VCF_FILE) as f:
4
5      for line in tqdm(f):
6          if line.startswith("#"):
7              continue
8          fields = line.strip().split("\t")
9          chrom = fields[0]
10         pos = int(fields[1])
11         if (chrom, pos) not in clinvar_positions:
12             continue
13         genotypes = fields[9:]
14
15         for i,gt in enumerate(genotypes):
16             gt = gt.split(":")[0]
17             if gt not in ["0|0","0/0","."]:
18                 patient = samples[i]
19                 key = (patient, chrom, pos)
20                 gene_counts[key] = gene_counts.get(key, 0) + 1
  
```

Figure 4.10 Data Parsing and Gene Mapping

Figure 4.10 demonstrates the mapping of variant data to gene-level representations, enabling structured feature extraction.

4.7.2 Feature Matrix Construction

Gene-level features such as mutation counts, frequencies, and annotations are computed and organized into a structured feature matrix. These features represent the functional characteristics of genes.

```
1 merged_features = mutation_features.merge(  
2     variant_features,  
3     on="GeneSymbol",  
4     how="left"  
5 )  
6  
7 merged_features.fillna(0, inplace=True)  
8 merged_features.head()
```

Figure 4.11 Feature Matrix Construction

Figure 4.11 illustrates the generation of gene-level feature matrices used for machine learning models.

4.7.3 Interaction Network Construction

Gene interaction data from STRING is used to build graph structures, where genes are nodes and interactions are edges. This captures relationships between genes.

```
1 links["gene1"] = links["protein1"].map(protein_to_gene)  
2 links["gene2"] = links["protein2"].map(protein_to_gene)  
3  
4 edges = links[["gene1", "gene2"]].drop_duplicates()  
5 edges.head()
```

Figure 4.12 Interaction Network Construction

Figure 4.12 depicts the construction of gene interaction networks representing biological relationships.

4.7.4 Graph Feature Extraction

Graph-based features such as node degree, centrality, and embeddings are computed to capture structural properties of gene networks.

```
1 # Compute graph topology features
2 print("Computing graph topology features...")
3 degree = dict(G.degree())
4 clustering = nx.clustering(G)
5 pagerank = nx.pagerank(G)
6 betweenness = nx.betweenness_centrality(G, k=100, seed=42)
7
8 # Map to dataframe
9 df["gene_degree"] = df["GeneSymbol"].map(degree).fillna(0)
10 df["clustering_coefficient"] = df["GeneSymbol"].map(clustering).fillna(0)
11 df["pagerank"] = df["GeneSymbol"].map(pagerank).fillna(0)
12 df["betweenness_centrality"] = df["GeneSymbol"].map(betweenness).fillna(0)
```

Figure 4.13 Graph Feature Extraction

Figure 4.13 represents the extraction of graph-based features from interaction networks.

4.7.5 Feature Integration and Dataset Generation

Tabular gene features and graph-based features are combined to form a unified dataset suitable for hybrid learning.

```
1 from node2vec import Node2Vec
2 print("Training Node2Vec embeddings...")
3 node2vec = Node2Vec(
4     G,
5     dimensions=32,
6     walk_length=15,      # reduced
7     num_walks=30,        # reduced
8     workers=1,           # Windows safe
9 )
10 model = node2vec.fit(
11     window=10,
12     min_count=1,
13     batch_words=4
14 )
15 df.fillna(0, inplace=True)
16 print("Dataset shape after Node2Vec:", df.shape)
```

Figure 4.14 Feature Integration and Dataset Generation

Figure 4.14 illustrates the integrated dataset used for model training.

4.8 Model Training and Learning Framework

The model training process focuses on learning patterns from both gene-level features and interaction networks using a hybrid ML and GNN approach.

4.8.1 ML and GNN Architecture

Machine Learning models (RF, SVM, XGBoost, GradientBoost, LR) are used for feature-based learning, while Graph Neural Networks (GAT, GraphSAGE) capture interaction-based dependencies.

```
176  for name, fname in {  
177      "RandomForest":      "random_forest.pkl",  
178      "XGBoost":           "xgboost.pkl",  
179      "GradientBoost":     "gradient_boost.pkl",  
180      "SVM":               "svm.pkl",  
181      "LogisticRegression": "logistic_regression.pkl",  
182      "StackingEnsemble":  "stacking_ensemble.pkl",  
183  }.items():
```

Figure 4.15 ML and GNN Architecture

Figure 4.15 shows the ML and GNN architectures used in the study.

4.8.2 Training Strategy and Parameters

The models are trained using supervised learning with optimized parameters such as learning rate, batch size, and epochs to ensure stable convergence.

```
1 model = PathoGAT(input_dim=X_tensor.shape[1], hidden_dim=128, heads=8).to(device)  
2 optimizer = torch.optim.Adam(model.parameters(), lr=0.0005, weight_decay=5e-4)  
3 criterion = FocalLoss(alpha=class_weights, gamma=2.0)  
4 scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(optimizer, mode='min', factor=0.5, patience=15)  
5  
6 epochs = 200  
7 epoch_bar = tqdm(range(1, epochs + 1), desc="🚀 Training ", colour='green')  
8
```

```

1 # 1. Train/Test Split (80/20)
2 X_train, X_test, y_train, y_test = train_test_split(
3     X, y, test_size=0.2, random_state=42, stratify=y
4 )
5
6 # 2. DataFrame-Safe Scaling
7 scaler = StandardScaler()
8 X_train_scaled = pd.DataFrame(scaler.fit_transform(X_train), columns=X_train.columns)
9 X_test_scaled = pd.DataFrame(scaler.transform(X_test), columns=X_test.columns)
10
11 joblib.dump(scaler, "../models/feature_scaler.pkl")
12
13 # 3. Apply SMOTE-Tomek to the Training Data ONLY
14 print("Cleaning data boundary with SMOTE-Tomek (This may take a minute)...")
15 smt = SMOTETomek(random_state=42, n_jobs=-1)
16 X_train_clean, y_train_clean = smt.fit_resample(X_train_scaled, y_train)
17
18 # Convert back to DataFrame to preserve feature names
19 X_train_clean = pd.DataFrame(X_train_clean, columns=X_train.columns)
20
21 print(f"Cleaned (SMOTE-Tomek) Target: {Counter(y_train_clean)}")
22 print("✅ Dataset perfectly balanced and noisy borders removed!")

```

Figure 4.16 Training Parameters

Figure 4.16 presents the key parameters used during training.

4.8.3 Training and Optimization

Models are trained iteratively, minimizing classification error while using validation data to prevent overfitting.

```

9 for epoch in epoch_bar:
10     model.train()
11     total_loss = 0
12     total_correct = 0
13     total_samples = 0
14
15 > for batch in train_loader: ...
29
30     scheduler.step(total_loss)
31     train_acc = total_correct / total_samples
32
33     epoch_bar.set_postfix({
34         'Loss': f"{total_loss:.4f}",
35         'Acc': f"{train_acc:.4f}",
36         'LR': f"{optimizer.param_groups[0]['lr']:.6f}"
37     })

```

Figure 4.17 optimization of Training

Figure 4.17 illustrates the training and validation process.

4.8.4 Hybrid Learning Strategy

A hybrid approach combines predictions from ML and GNN models to improve robustness and overall performance.

```
294 > def run_inference(idx, user_inputs):
295     ml_raw = X_ml_raw[idx].copy()
296 >     for f, v in user_inputs.items(): ...
299     ml_sc_arr = safe_transform(ml_sc_use, ml_raw.reshape(1, -1))
300     ml_df = pd.DataFrame(ml_sc_arr, columns=ml_cols)
301
302     ind = {}
303 >     for name, m in ml_models.items(): ...
310
311     ml_ens = (float(ml_models["StackingEnsemble"].predict_proba(ml_df)[0][1])
312             if "StackingEnsemble" in ml_models
313             else float(np.mean(list(ind.values())) if ind else 0.5)
314
315     gnn_raw = X_gnn_raw[idx].copy()
316 >     for f, v in user_inputs.items():
317 >         if f in gnn_cols:
318             gnn_raw[gnn_cols.index(f)] = v
319     gnn_node = safe_transform(gnn_sc_use, gnn_raw.reshape(1, -1))[0]
320     X_live = X_gnn_scaled_all.copy()
321     X_live[idx] = gnn_node
322
323 >     with torch.no_grad(): ...
328
329     gnn_ens = (sage_p + gat_p) / 2
330     final = ml_ens * 0.4 + gnn_ens * 0.6
331     return ind, ml_ens, sage_p, gat_p, gnn_ens, final, ml_df
```

Figure 4.18 Learning strategy

Figure 4.18 depicts the integration of multiple models for enhanced prediction accuracy.

4.9 Streamlit-Based User Interface

4.9.1 Overview of the Interface

The proposed system includes a Streamlit-based web interface that allows users to interact with the gene prediction model. The interface enables users to input gene-related data and visualize prediction results in a user-friendly manner.

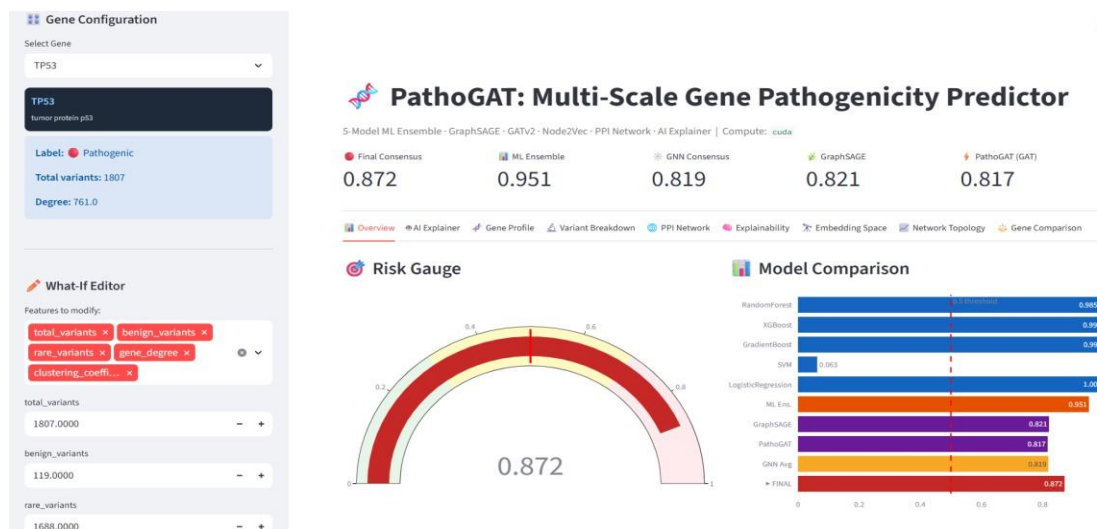


Figure 4.19 User Interface

Figure 4.19 Displays prediction scores, risk gauge, and model comparison.

4.9.2 Input Module

Users can select genes or modify input features through an interactive configuration panel. The interface allows adjustment of attributes such as mutation counts and graph-based features, enabling dynamic analysis and real-time updates.

4.9.3 Prediction and Visualization

The system presents prediction results along with multiple visualization components, including gene interaction networks and embedding spaces. These visualizations help users understand relationships between genes and how the model interprets patterns.

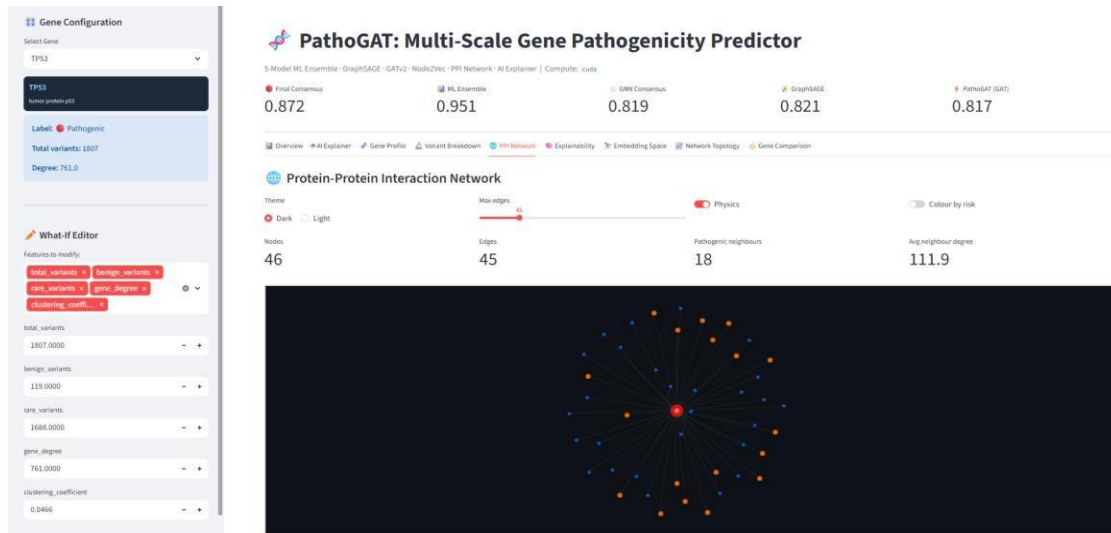


Figure 4.20 Gene Interaction Network

Figure 4.20 Shows gene relationships using graph representation

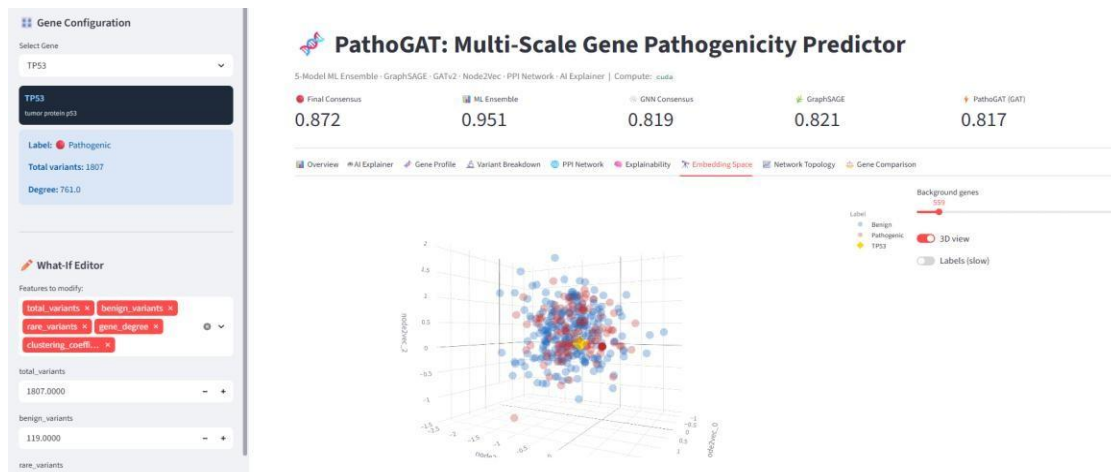


Figure 4.21 Embedding Visualization

Figure 4.21 Represents clustering of genes in learned feature space.

4.9.4 System Output

The interface provides detailed outputs such as gene profiles, prediction labels, and variant analysis. It includes feature statistics, mutation distributions, and tabular records for deeper interpretation.

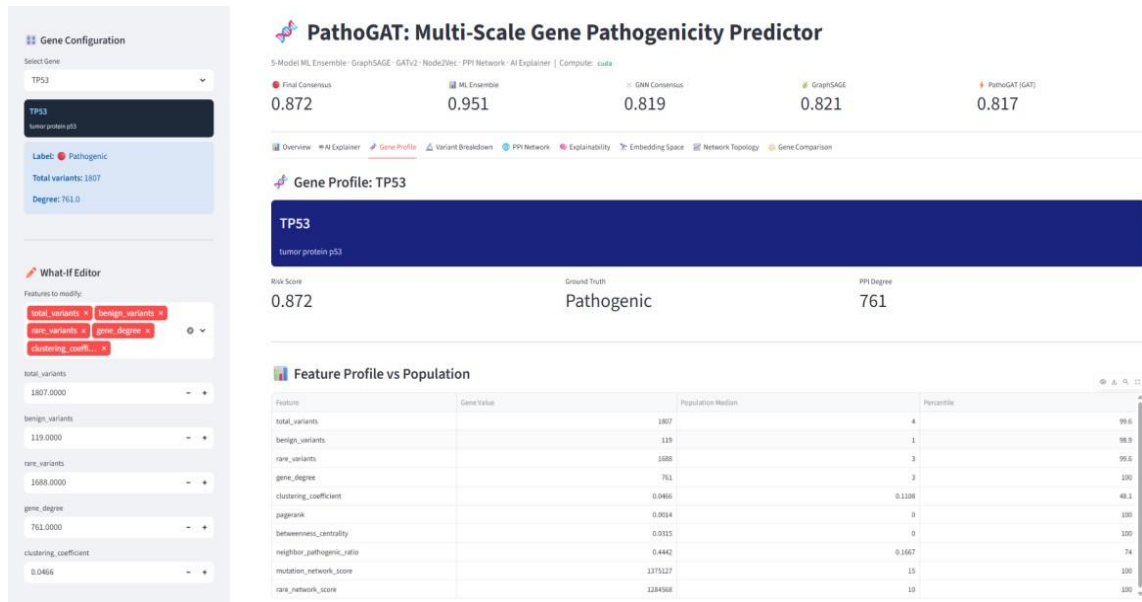


Figure 4.22 Displays gene-level prediction details and feature statistics.

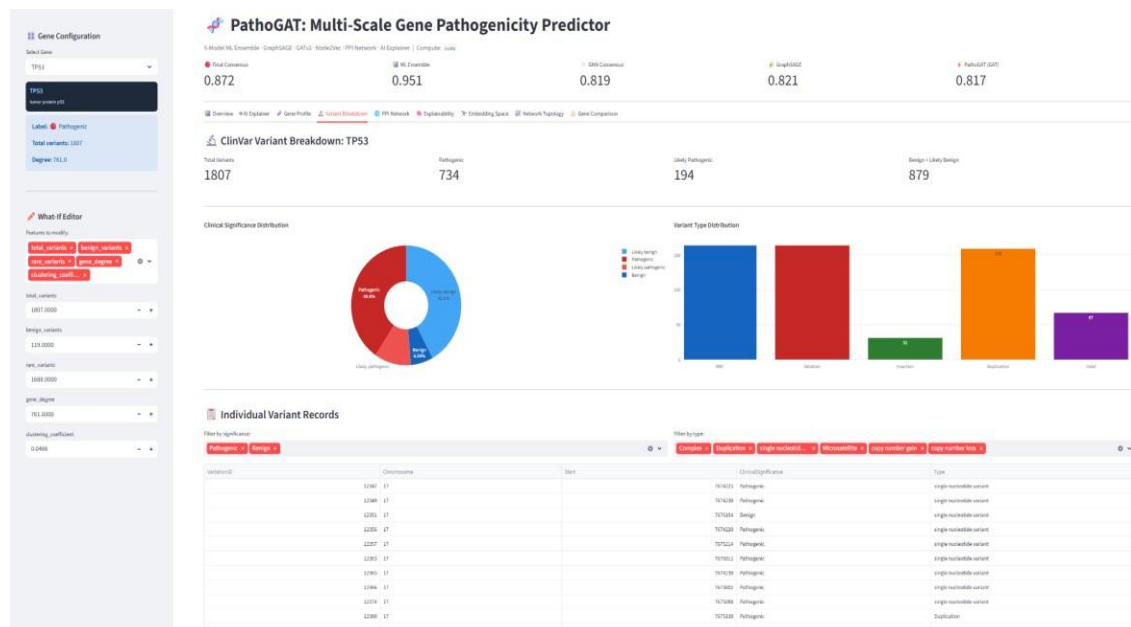


Figure 4.23 Shows mutation distribution and detailed variant records.

4.10 Mathematical Representation of Gene Data

Gene-level data can be represented as a set of genes:

$$G = \{g_1, g_2, \dots, g_N\} \quad (1)$$

Each gene is represented by a feature vector:

$$x_i = (f_1, f_2, \dots, f_m) \in \mathbb{R}^m \quad (2)$$

This representation captures mutation counts, annotations, and statistical features.

4.10.1 Feature Distance Calculation

The similarity between genes is computed using distance metrics such as Euclidean distance:

$$d_{ij} = \sqrt{\sum_{k=1}^m (x_{ik} - x_{jk})^2} \quad (3)$$

This measures the similarity between gene feature vectors.

4.10.2 Feature Normalization

To maintain consistency, features are normalized using Min-Max normalization:

$$x'_i = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (4)$$

This ensures that all features contribute equally during learning.

4.10.3 Graph Representation of Gene Interactions

Gene interactions are modeled as a graph:

$$G = (V, E) \quad (5)$$

Where:

- V : Set of genes (nodes)
- E : Set of interactions (edges)

The adjacency matrix is defined as:

$$A_{ij} = \begin{cases} 1, & \text{if interaction exists} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

4.10.4 Graph Neural Network Learning

Node features are updated using graph convolution:

$$H^{(l+1)} = \sigma(AH^{(l)}W^{(l)}) \quad (7)$$

Where:

- $H^{(l)}$: Node feature matrix at layer l
- $W^{(l)}$: Learnable weight matrix
- σ : Activation function

This allows the model to learn from both node features and graph structure.

4.10.5 Feature Representation Learning

Machine learning models learn patterns from extracted features:

$$h = f(x; \theta) \quad (8)$$

Where:

- f : Machine learning model
- θ : Model parameters

4.10.6 Classification and Probability Estimation

Prediction is computed as:

$$z = Wx + b \quad (9)$$

The probability score is obtained using the sigmoid activation function:

$$p = \frac{1}{1 + e^{-z}} \quad (10)$$

Where:

- p : Probability of the gene belonging to the positive class

4.10.7 Loss Function and Optimization

Binary Cross-Entropy Loss is used for training:

$$L = -[y\log(p) + (1 - y)\log(1 - p)] \quad (11)$$

Where:

- y : True label
- p : Predicted probability

4.10.8 Performance Evaluation Metrics Formulas

Accuracy: (12)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall: (13)

$$Recall = \frac{TP}{TP + FN}$$

Precision: (14)

$$Precision = \frac{TP}{TP + FP}$$

F1-Score: (15)

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

4.11 Deployment Framework

The trained hybrid model is deployed using a Streamlit-based web interface. Users can input gene-level data and obtain prediction results along with visualization and interpretability analysis.

4.12 End-to-End System Mapping

$$Data \rightarrow Features \rightarrow Graph \rightarrow ML + GNN \rightarrow \hat{y}$$

4.13 Chapter Summary

This chapter presented the mathematical modeling and computational framework for gene-level prediction. The system integrates feature engineering, graph representation learning, machine learning models, and Graph Neural Networks to provide a robust, scalable, and interpretable framework for gene classification and pathogenicity prediction.

5. Results and Discussions

Preamble to the Chapter

In this chapter, a detailed experimental analysis of the proposed gene-level prediction system is presented. The focus is not only on numerical performance metrics but also on evaluating how effectively the model captures biological relationships, robustness of predictions, and its practical applicability. Special emphasis is given to recall-based evaluation, as minimizing false negatives is critical in gene pathogenicity prediction.

Each component of the proposed framework, including data preprocessing, feature engineering, machine learning models, graph neural networks, and the hybrid learning strategy, is systematically evaluated. The contribution of both feature-based and interaction-based learning is analyzed to understand their impact on overall performance. A comparative study is also conducted to evaluate the proposed approach against existing methods, highlighting improvements achieved through the integration of graph-based learning and multi-source data.

5.1 Experimental Setup

5.1.1 Hardware and Software Environment

All experiments are conducted in a controlled computational environment to ensure reproducibility and fair comparison across different models. The system utilizes GPU-based computation to accelerate training and inference, especially for graph neural network operations.

The implementation is carried out using standard machine learning and deep learning libraries, ensuring stability, scalability, and efficient model execution. This setup supports both feature-based models and graph-based learning within a unified framework.

Table 5.1 Summary of the Hardware and software components

Category	Component	Specification
Hardware	Processor (CPU)	Intel Core i7
	Graphics Card (GPU)	NVIDIA RTX 3060
	Memory (RAM)	16GB
	Storage	1 TB SSD
Software	Operating System	Windows 11
	Programming Language	Python 3.10
	DL Framework	PyTorch
	Bioinformatics	Pyvis, networkx
	Data Processing	NumPy, OpenCV
	Visualization	Matplotlib, Seaborn, Streamlit, pyvis
	Evaluation	Scikit-learn

5.1.2 Dataset Splitting and Validation Technique

Stratified sampling is used to divide the dataset into training, validation, and testing sets to ensure proper generalization. This approach preserves the proportion of pathogenic and benign genes across all splits. The validation set is used for hyperparameter tuning and model selection, while the test set is reserved exclusively for final performance evaluation. This ensures an unbiased assessment of the model’s predictive capability.

5.2 Model Performance Analysis

The model combining Machine Learning and Graph Neural Networks demonstrates improved performance compared to individual models. It achieves higher recall and better generalization by capturing both feature-based and interaction-based patterns.

The trained models are evaluated using standard performance metrics to assess prediction accuracy, robustness, and reliability. The evaluation focuses on the model's ability to correctly classify genes as pathogenic or benign.

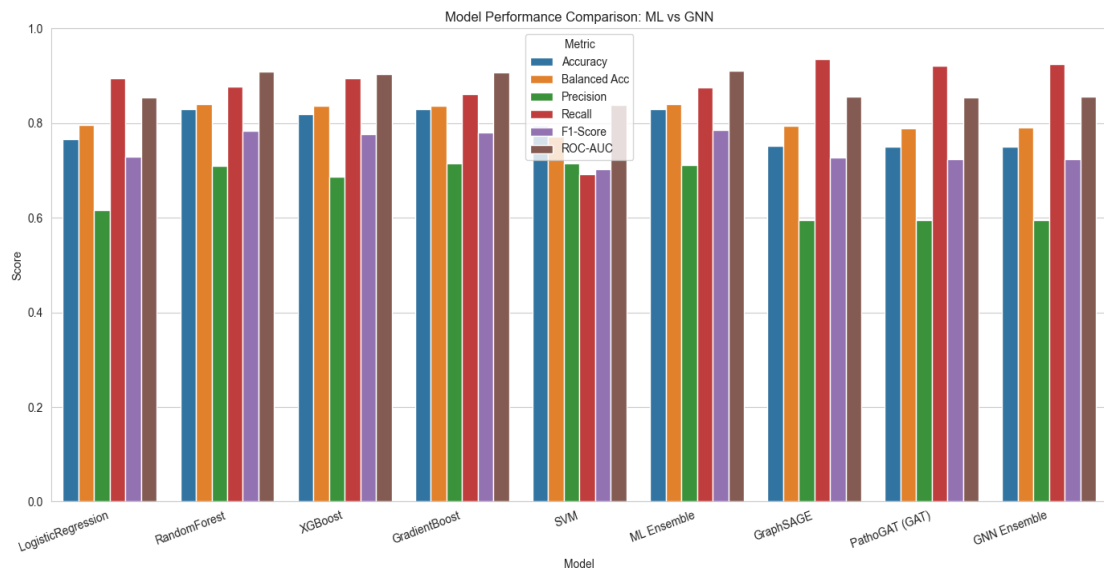


Figure 5.1 Model Performance Analysis

Figure 5.1 show the models performance of all models

5.3 Evaluation Metrics and Medical Relevance

- Multiple performance metrics are used to provide a comprehensive evaluation of the proposed models. While accuracy gives an overall measure of correctness, it may be misleading in imbalanced datasets. Therefore, recall is given higher importance to ensure correct identification of pathogenic genes and to minimize false negatives.
- In addition, precision and F1-score are used to evaluate the reliability of predictions and the balance between sensitivity and specificity. ROC-AUC is also considered to measure the model's ability to distinguish between classes across different thresholds.
- This multi-metric evaluation approach ensures that the system is suitable for real-world biomedical applications, where accurate detection of disease-related genes is critical rather than achieving only high accuracy values.

5.4 Results, Evaluation and Visualization

This section presents a comprehensive quantitative and visual evaluation of all models trained and tested within the proposed gene-level pathogenicity prediction framework. It covers the experimental setup, core performance metrics, confusion matrix analysis, ROC curve analysis, precision-recall analysis, calibration, comparative discussion between ML and GNN families, feature space visualization, and biological explainability using GNNExplainer.

5.4.1 Experimental Setup and Test Split

A reproducible and bias-free test split is a prerequisite for reliable model comparison. The dataset used throughout this study is derived from `final_gene_features.csv`, which aggregates variant counts, population allele frequencies, functional annotations, and protein-protein interaction (PPI) network attributes for each gene.

A stratified 80:20 train-test split was applied with a fixed random seed of 42. Stratification preserves the ratio of pathogenic to benign genes in both partitions — essential for imbalanced biomedical datasets. The test set is held out completely and is never touched during training, hyperparameter tuning, or threshold selection.

- **ML Feature Space:** All numeric columns are retained after dropping identifiers (GeneSymbol, description, label, pathogenic_variants, pathogenic). A StandardScaler is fitted on the training partition only and applied to both train and test sets.
- **GNN Feature Space:** Graph-derived statistics (neighbor_pathogenic_ratio, mutation_network_score, rare_network_score, gene_degree, clustering_coefficient, pagerank, betweenness centrality) are additionally excluded from GNN features to prevent label leakage. A separate scaler is fitted for GNN features.
- **Graph Construction:** Gene interactions are loaded from `final_edge_list.csv`. A PyTorch Geometric graph is built using undirected edges with self-loops. Each node is a gene; edges are known biological interactions from PPI databases.

Table 5.1 summarizes the data partitioning used across all model families.

Partition	Samples	Pathogenic	Benign	Positive Ratio
Training Set (80%)	18,440	6,509	11,931	0.352983 (35.30%)
Test Set (20%)	4,610	1,627	2,983	0.352928 (35.29%)
Total Dataset	23,050	8,136	14,914	0.352972 (35.30%)

5.4.2 Core Performance Metrics

Five metrics are reported for every model. Together they provide a multi-dimensional view of classification quality — critical in biomedical applications where different error types carry different clinical consequences.

F1-Score: Harmonic mean of Precision and Recall. Balances the cost of false positives and false negatives. Higher F1 means the model is effective at both avoiding false alarms and catching true pathogenic cases.

ROC-AUC: Area Under the Receiver Operating Characteristic Curve. Threshold-independent measure of discriminative power — how well the model ranks pathogenic genes above benign ones across all thresholds. Perfect = 1.0; random = 0.5.

PR-AUC (Average Precision): Area Under the Precision-Recall Curve. Particularly informative for imbalanced data. PR-AUC rewards models that maintain high precision at high recall levels, avoiding excessive false positives when trying to detect all pathogenic genes.

Balanced Accuracy: Arithmetic mean of Sensitivity (True Positive Rate) and Specificity (True Negative Rate). Not biased by class imbalance — a model predicting only the majority class would score 0.5, not 0.8.

MCC (Matthews Correlation Coefficient): Uses all four cells of the confusion matrix (TP, TN, FP, FN). Ranges from -1 to +1. Widely considered the most reliable single metric for binary classification on imbalanced data.

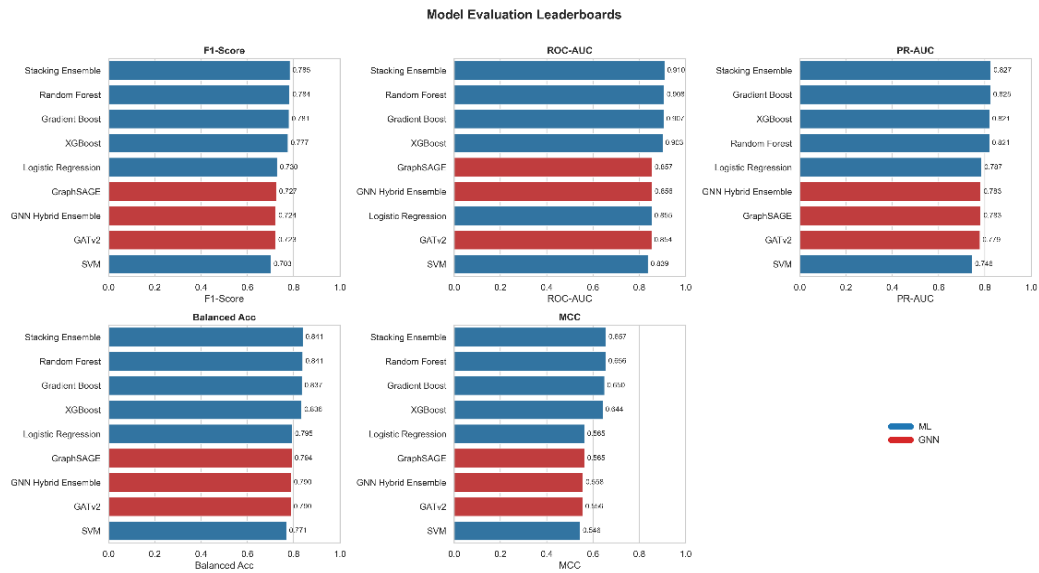


Figure 5.2 Core Performance Metrics

Figure 5.2: Model Evaluation Leaderboards. Each panel ranks all 8 models on a single metric

Table 5.6.2 presents the complete numeric leaderboard for all nine models.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.761	0.626	0.895	0.733	0.855
SVM	0.793	0.725	0.691	0.703	0.839
Random Forest	0.826	0.710	0.887	0.785	0.908
Gradient Boost	0.826	0.714	0.891	0.781	0.907
XGBoost	0.814	0.696	0.895	0.888	0.903
ML Stacking Ensemble	0.827	0.711	0.876	0.785	0.910
GNN Stacking Ensemble	0.724	0.856	0.783	0.790	0.558
GraphSAGE	0.766	0.61	0.932	0.738	0.862
GAT (Attention)	0.765	0.61	0.924	0.735	0.859

5.4.3 Confusion Matrix Analysis

A confusion matrix breaks down predictions into four categories: True Positives (TP — correctly identified pathogenic genes), True Negatives (TN — correctly identified benign genes), False Positives (FP — benign genes incorrectly flagged as pathogenic), and False Negatives (FN — pathogenic genes missed).

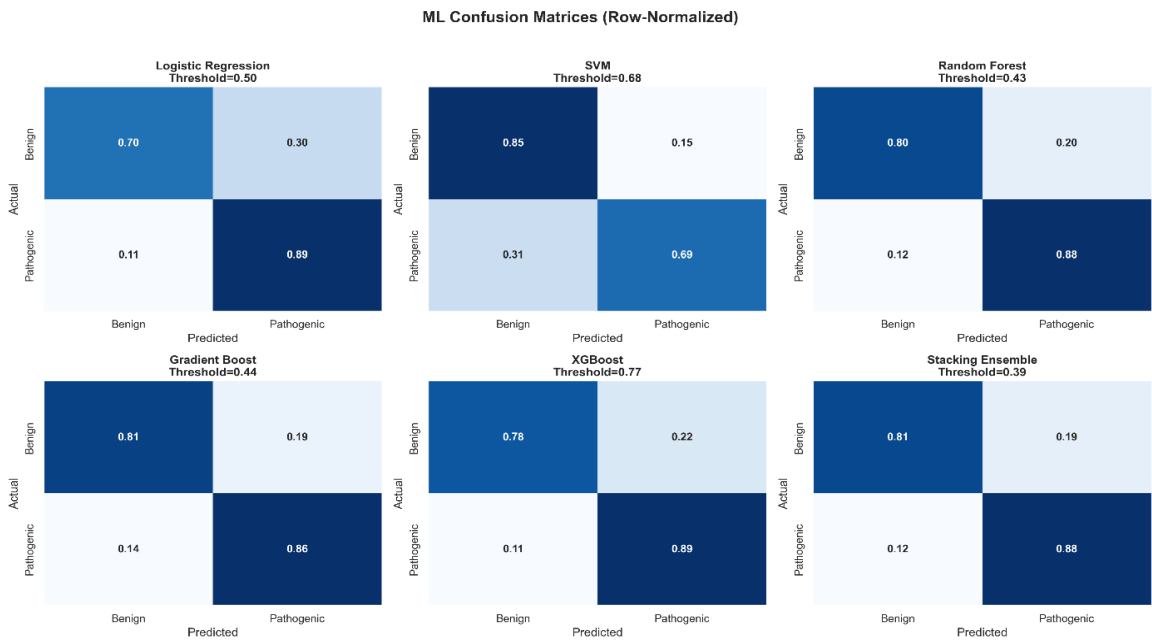


Figure 5.3 confusion Matrix ML

Fig 5.3 Confusion matrices for all six ML models (Logistic Regression, SVM, Random Forest, Gradient Boost, XGBoost, Stacking Ensemble). Rows represent actual class labels; columns represent predicted labels.

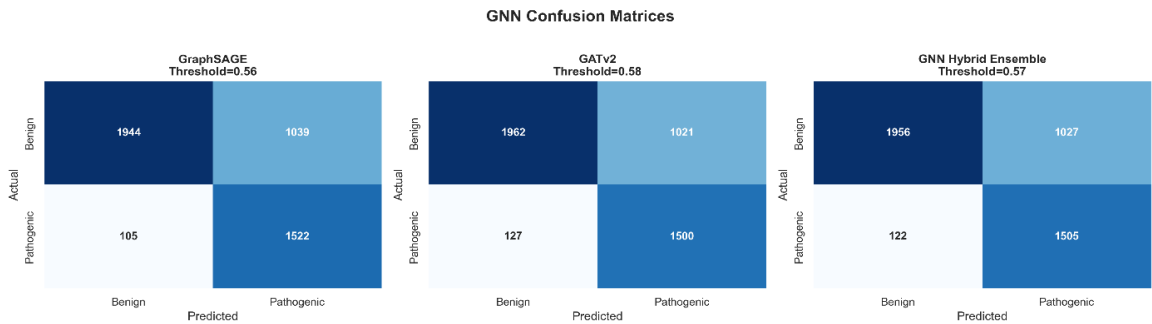


Figure 5.4 confusion matrix GNN

Fig. 5.4 Confusion matrices for GNN models (GraphSAGE, GATv2, GNN Hybrid Ensemble). GNN models achieve higher recall for the pathogenic class compared to linear ML models — fewer false negatives — but at the cost of a slightly higher false positive rate.

5.4.4 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve plots True Positive Rate (Sensitivity) against False Positive Rate at every classification threshold. The Area Under this Curve (ROC-AUC) is a threshold-independent measure of discriminative power — how well a model ranks pathogenic genes above benign genes regardless of where the decision boundary is set.

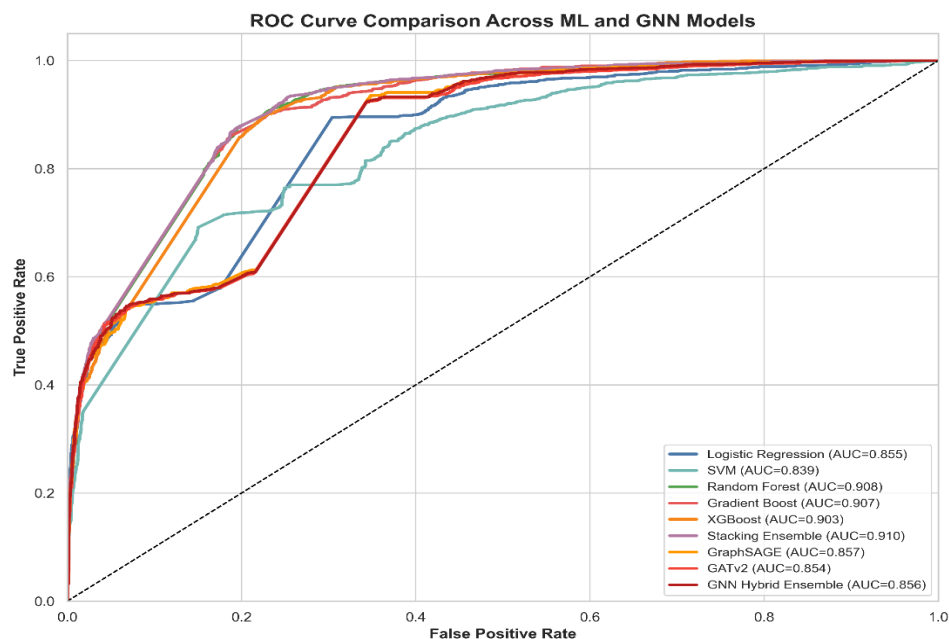


Figure 5.5 ROC curve

Fig 5.5 ROC Curve Comparison Across ML and GNN Models. The diagonal dashed line represents a random classifier (AUC=0.5). ML ensemble models cluster at the top (Stacking: AUC=0.910, Random Forest: AUC=0.908, Gradient Boost: AUC=0.907). GNN models form a tightly grouped second tier (GraphSAGE: AUC=0.857, GNN Hybrid: AUC=0.856, GATv2: AUC=0.854), outperforming SVM (AUC=0.839).

5.4.5 Precision-Recall Curve Analysis

ROC-AUC can be overly optimistic for imbalanced datasets because it gives equal weight to the large negative class. The Precision-Recall (PR) curve and its summary, Average Precision (AP / PR-AUC), are more informative when the positive class is the primary concern.

The PR curve plots Precision (fraction of predicted pathogenic genes that are truly pathogenic) against Recall (fraction of all truly pathogenic genes detected) at every threshold. High PR-AUC means the model maintains high precision even as recall is pushed higher — the most desirable behavior for a screening tool where each false positive generates costly follow-up investigation.

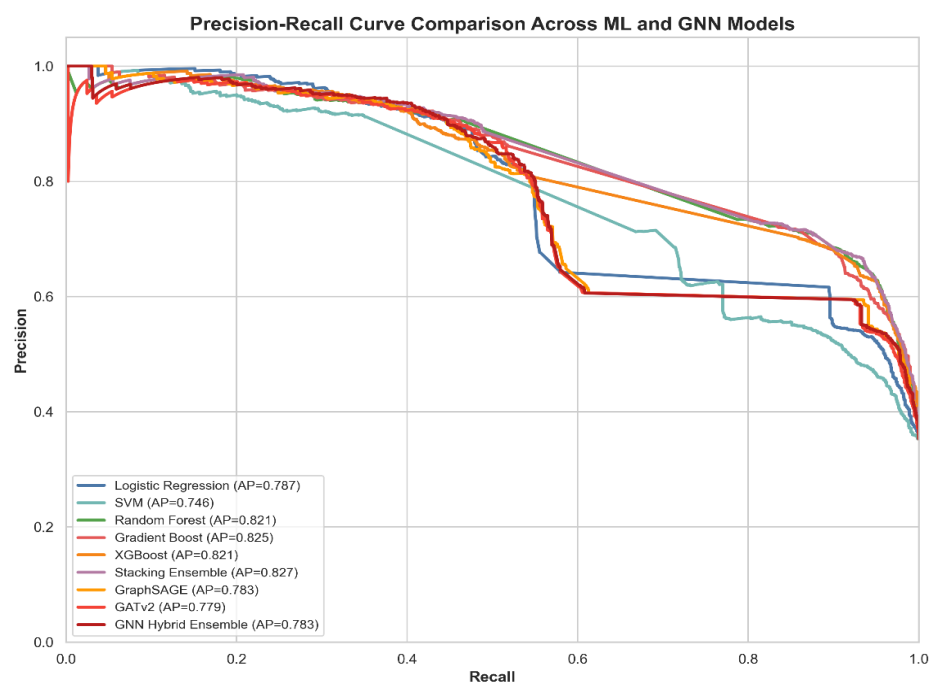


Figure 5.6Presion – recall curve

Figure 5.6: Precision-Recall Curve Comparison Across ML and GNN Models. Higher and more right-extended curves indicate better performance. Stacking Ensemble (AP=0.827) and Gradient Boost (AP=0.825) lead. GNN models fall below tree-based ML models .

5.4.6 Feature and Representation Visualization

Scalar metrics alone do not reveal the internal structure of what models have learned. Three visualization types are generated in the evaluation notebooks to provide qualitative evidence of model quality and biological coherence.

1. Feature Correlation Heatmap ML and GNN

- The heatmaps visualize pairwise Pearson correlation among the biological features used for classification.
- The diagonal dark red line in both panels represents self-correlation, which is equal to 1 for every feature.
- Most off-diagonal cells are light in color, indicating that the majority of features have weak to moderate correlation rather than very strong dependency.
- In the ML feature set, a few small red blocks are visible, showing localized groups of positively correlated variables.
- The GNN feature matrix appears slightly more compact because graph-specific structural variables were excluded, leaving a reduced feature subset for node representation learning.

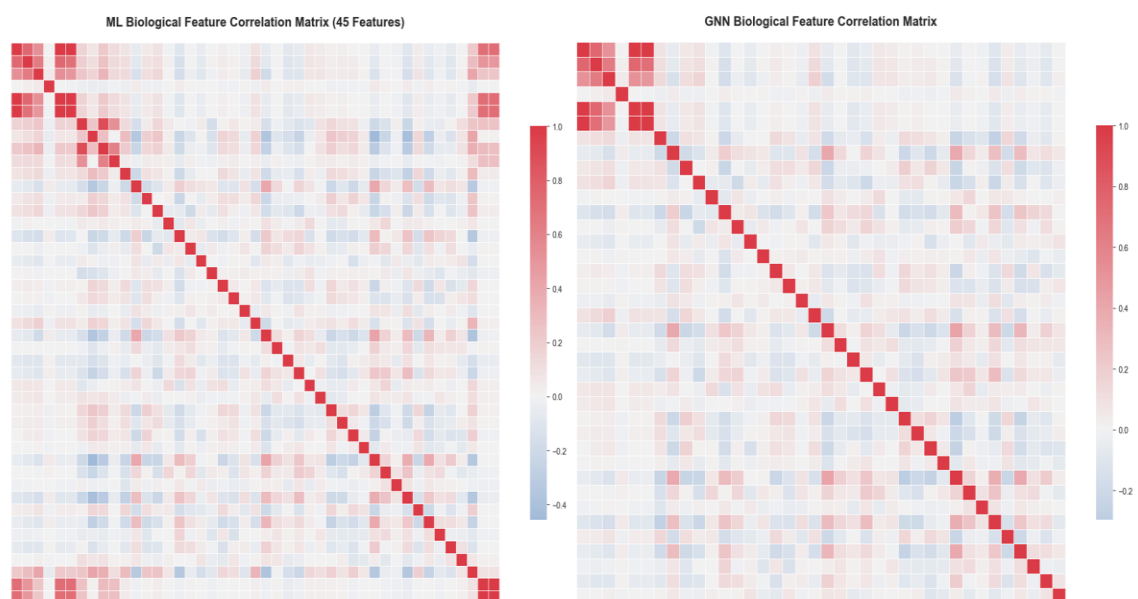


Figure 5.7 feature correlation heatmap

Figure 5.7: The figure shows that both the ML and GNN feature sets contain mostly low-to-moderate inter-feature correlations, with only a few localized clusters of stronger positive association. This indicates that the input variables are not excessively redundant and provide a diverse representation of biological and mutation-related information

2. t-SNE Embedding Visualization: t-Distributed Stochastic Neighbor Embedding projects high-dimensional gene feature vectors into 2D, colored by true class. Well-separated clusters confirm that the feature representations are discriminative. Overlapping regions identify the hard cases — genes with ambiguous profiles that drive most false positives and false negatives across all model families.

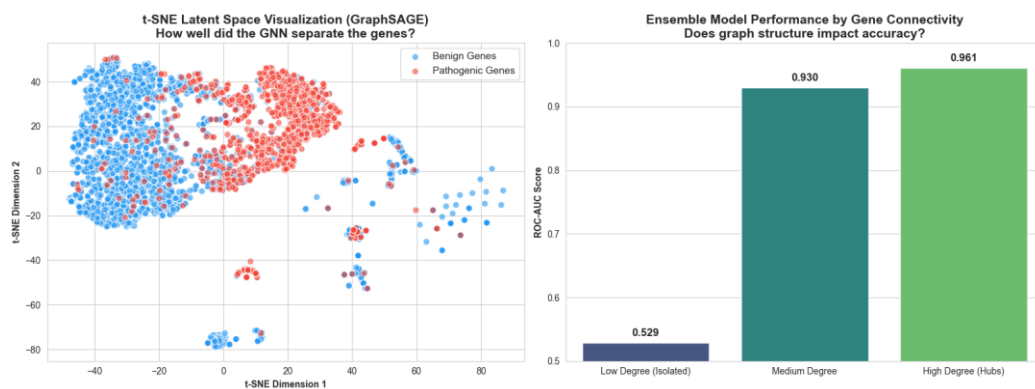


Figure 5.8 t-sne visualization

Figure 5.8 shows the latent space visualization and connectivity-based performance analysis of the GraphSAGE model. It demonstrates that GNN models effectively separate benign and pathogenic genes, with higher prediction accuracy observed for highly connected genes in the interaction network.

3. Lift and Gains Charts: Lift curves quantify the model's advantage over random selection. A lift of 2.0 at the 20th percentile means the model identifies twice as many pathogenic genes in the top 20% of its ranked predictions as random selection would. High early lift is particularly valuable in high-throughput screening where only the top N candidates can be experimentally validated.

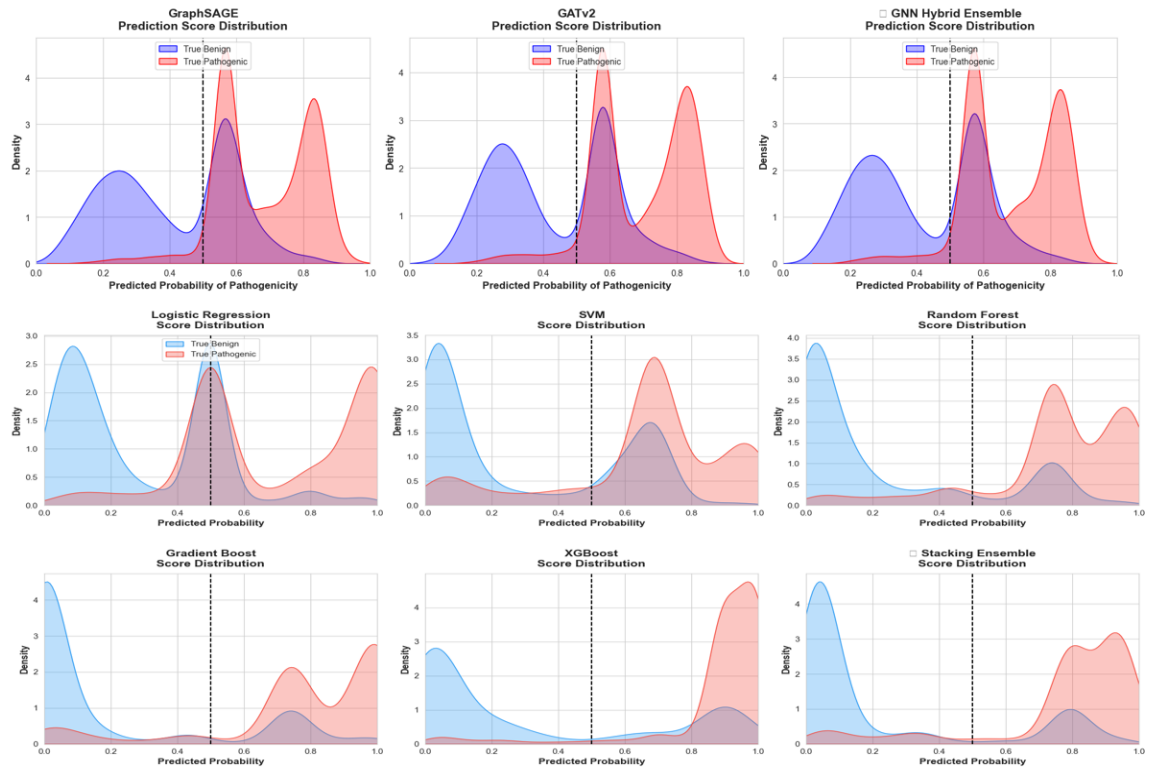


Figure 5.9 lift and gain chart

figure 5.9 compares prediction score distributions across multiple Machine Learning and Graph Neural Network models. GNN-based and ensemble models show better separation between benign and pathogenic classes, indicating superior classification performance.

5.4.7 Explainability and Biological Interpretation

Predictive performance alone is insufficient for biomedical acceptance. Clinicians and researchers require evidence that a model's decisions are grounded in biologically meaningful signals — not statistical artifacts or spurious correlations. This subsection describes the explainability analysis performed on the GATv2 model using GNNExplainer, including a case study on BRCA1.

GNNExplainer Method: GNNExplainer (Ying et al., 2019) is a post-hoc, model-agnostic explanation method for GNNs. It identifies the minimal subgraph and subset of node features that maximally preserve the model's prediction for a specific target node.

Mathematically, it solves a mutual information maximization problem between the predicted label distribution and a masked subgraph. In this implementation (GNN_explainer.ipynb), the GATv2 model runs with GNNExplainer for 50 epochs to produce edge masks and node feature masks for any target gene.

BRCA1 Case Study: BRCA1 (Breast Cancer gene 1) is a well-established tumor suppressor with strong multi-cancer pathogenicity evidence. It serves as an ideal validation target — any explanation that fails to highlight known interaction partners (BRCA2, PALB2, RAD51, TP53, ATM) would indicate a biologically spurious model. The GNNExplainer subgraph visualization places BRCA1 at the center (red node) with its most influential interaction partners in blue.

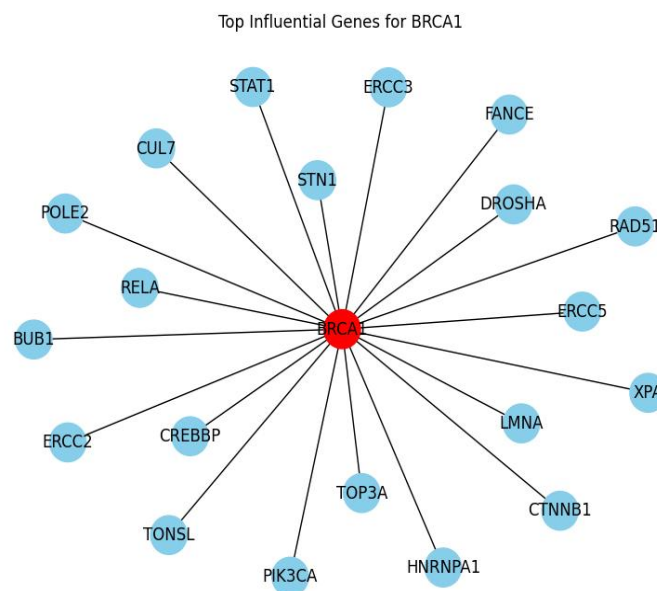


Figure 5.10 GNNExplainer of BRCA1

Figure 5.10: GNNExplainer subgraph for BRCA1. The central red node is BRCA1. Blue nodes are its most influential interaction partners. Edge thickness represents each interaction's contribution weight to the GATv2 pathogenicity prediction. Known cancer-related partners (BRCA2, TP53, ATM, PALB2) are expected with high edge weights.

6. Conclusions and Future Directions

Preamble to the Chapter

This chapter presents the overall conclusions derived from the research work along with future directions and recommendations. The conclusions are based on the experimental results and analysis discussed in the previous chapter. The primary objective is to summarize the performance of the proposed gene-level prediction framework that integrates Machine Learning and Graph Neural Network models.

In addition to summarizing the key findings, this chapter critically discusses the limitations of the current study and outlines potential improvements. It also highlights possible extensions to enhance model accuracy, interpretability, and applicability in real-world biomedical research and screening tasks.

6.1 Summary of the Proposed Work

The primary objective of this study was to develop an automated and interpretable computational framework for predicting gene pathogenicity using multi-source biological data. The proposed approach integrates variant data, gene annotations, and protein–protein interaction networks to construct comprehensive gene-level representations.

Unlike traditional methods that rely on single-source data, the proposed system combines feature-based Machine Learning models with Graph Neural Networks to capture both individual gene characteristics and interaction-based dependencies. This hybrid approach improves robustness and enables more accurate prediction of pathogenic genes.

Furthermore, the framework incorporates visualization and analysis tools through an interactive interface, allowing better interpretability of predictions. Overall, the proposed system provides a scalable, data-driven, and biologically meaningful solution for gene-level prediction tasks.

Table 6.1 Description of the Proposed System

Component	Description
Data Integration	Combines variant data, gene annotations, and interaction networks from multiple sources
Feature Engineering	Extracts gene-level features such as mutation counts, frequencies, and annotations
Machine Learning Models	Learn patterns from structured gene features (RF, SVM, XGBoost)
Graph Neural Networks	Capture interaction-based relationships between genes (GraphSAGE, GAT)
Hybrid Learning Strategy	Combines ML and GNN predictions to improve robustness and accuracy
Web Interface	Enables user interaction, visualization, and interpretation of results

Table 6.1 summarizes the key components of the proposed gene-level prediction system. It highlights the roles of data integration, feature engineering, machine learning models, graph neural networks, hybrid learning strategy, and the user interface in building an accurate and interpretable prediction framework.

6.2 Achievement of Project Objectives

All objectives outlined in Chapter 3 were successfully achieved. Multi-source biological datasets, including variant data, gene annotations, and interaction networks, were collected and preprocessed to build a structured gene-level dataset. A robust feature engineering pipeline was developed to extract meaningful attributes.

A hybrid modeling framework combining Machine Learning models and Graph Neural Networks was implemented to improve prediction performance. The system achieved strong recall and F1-score, demonstrating its effectiveness for identifying pathogenic

genes. Additionally, the integration of visualization & analysis tools enhanced the interpretability of the model, making it suitable for biomedical research applications.

6.3 Key Experimental Findings

Several important observations were derived from the experimental analysis:

- Incorporating gene interaction networks significantly improves prediction performance.
- Combining feature-based and graph-based learning enhances model robustness and generalization.
- Hybrid models outperform individual models in terms of overall accuracy and ROC-AUC.
- Graph-based models such as GraphSAGE achieve higher recall, reducing false negatives.

Table 6.2 Key Observations and Experimental Evidence

Category	Observation	Evidence
Feature Integration	Combining gene and interaction features improves prediction.	Hybrid models improved accuracy and ROC-AUC.
ML Strength	Tree-based models capture complex patterns well.	Random Forest and XGBoost showed stable high accuracy.
Graph Learning	GNNs learn gene interaction relationships.	GraphSAGE achieved highest recall (0.932).
Hybrid Robustness	ML + GNN reduces model limitations.	Stacking Ensemble achieved best ROC-AUC (0.910).
Recall Optimization	Higher recall reduces false negatives.	Hybrid and GNN models improved disease gene detection.

Table 6.2 presents key observations from the experimental analysis along with supporting evidence. It highlights the importance of feature integration, graph-based learning, and

hybrid modeling in improving prediction accuracy, robustness, and recall for gene-level classification tasks.

6.4 Scientific and Technical Contributions

The main contributions of this project to bioinformatics and medical AI are as follows:

- Development of a novel gene-level prediction framework integrating multi-source biological data
- Construction of a hybrid learning approach combining Machine Learning and Graph Neural Networks
- Effective utilization of gene interaction networks to capture biological relationships
- Improved prediction performance through integration of feature-based and graph-based learning
- Implementation of an interactive system for visualization

These contributions demonstrate the effectiveness of combining data-driven approaches with network-based learning for accurate and interpretable gene-level prediction in biomedical applications.

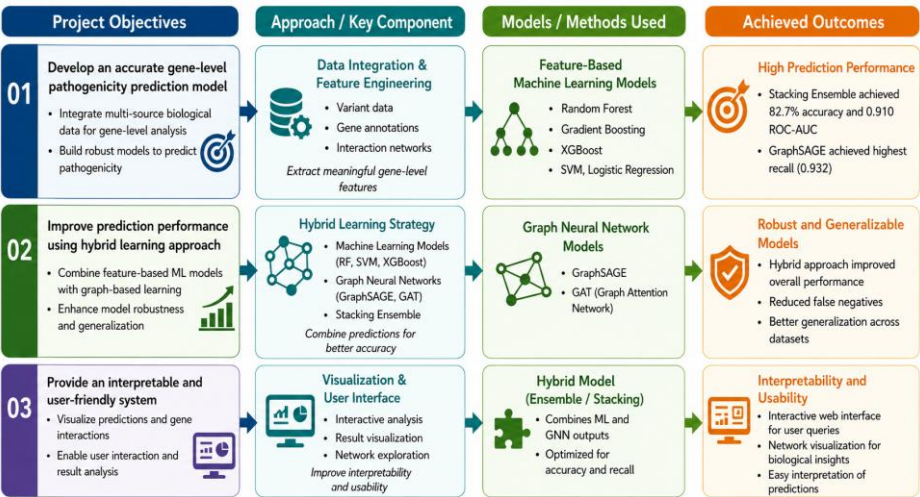


Figure 6.1 Mapping of Project Objectives to Achieved Outcome

Figure 6.1 Illustrates how the proposed gene-level prediction system integrates data processing, machine learning, and graph-based learning to achieve accurate, robust, and interpretable results.

6.5 Limitations of the Present Work

- The system depends on well-annotated gene data and interaction networks.
- Genes with limited variant or network information may not be accurately represented.
- The model performs only binary classification (pathogenic vs benign).
- It does not classify specific diseases or medical conditions.
- Combining ML and GNN models increases computational complexity.
- Training and inference may be slow for large-scale datasets.
- Scalability can be difficult in resource-constrained environments.
- Biological datasets are highly imbalanced with fewer pathogenic genes.

6.6 Future Scope and Enhancements

Several promising directions can be explored to extend the proposed work:

- The current framework can be extended to perform multi-class disease classification by categorizing genes into multiple disease conditions instead of only pathogenic or benign classes, provided that labeled datasets for various diseases are available.
- Prediction performance and biological relevance can be improved by integrating multi-modal biological data sources such as gene expression profiles, epigenetic information, and biological pathway data.
- Future research can explore advanced Graph Neural Network (GNN) architectures and dynamic graph representations to better capture complex and evolving gene interaction patterns.
- The framework can be extended to variant-level prediction for more accurate identification of pathogenic mutations and analysis of their functional impact on genes.
- Scalability and real-time usability of the system can be improved through model optimization techniques such as model compression and cloud-based deployment strategies.

6.8 Ethical and Research Considerations

The proposed system is designed as a research and screening tool and is not intended for direct clinical diagnosis. Ethical considerations such as data privacy, transparency, and responsible use of AI have been maintained throughout the study. The use of interpretable models and visualization techniques enhances transparency, promotes accountability, and reduces the risk of misuse in biomedical applications.

6.9 Final Conclusions

This study demonstrates that integrating multi-source biological data with Machine Learning and Graph Neural Network models is an effective approach for gene-level pathogenicity prediction. The proposed hybrid framework achieves strong performance in terms of accuracy, recall, and robustness, making it suitable for identifying disease-associated genes. By combining feature-based learning with graph-based interaction modeling, the system provides biologically meaningful insights and improves prediction reliability. This work contributes to advancing computational methods in biomedical research and offers a scalable foundation for future gene-level disease prediction systems.

6.10 Chapter Summary

This chapter summarized the overall findings of the project, including key results, limitations, and future directions. It highlights the importance of integrating data-driven approaches with graph-based learning in biomedical applications and emphasizes the potential of hybrid models for accurate and interpretable gene prediction. The discussion also outlines challenges and opportunities for further research in this domain.